

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

Generated from `docs/traceability.toml`

Overview

Implemented	Descriptive	Implicit	Out-of-scope
86	33	2	3

Total mappings: 124 · Total impl sites: 266

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 1/3 implemented

§1 – Introduction

§1 – Introduction

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Introduction

§1.1 – General Comments

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General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:195 GH:omdurman-rules/src/unit_profiles.rs:195	command_owns_unit	<pre>193 /// treat *nobody-claims-it* units as the faction's communal pool, so `false` 194 /// here means "any member of the faction may act on this unit". 195 pub fn command_owns_unit(scope: &omdurman_types::CommandScope, identity: &UnitIdentity) -> bool { 196 match scope { 197 omdurman_types::CommandScope::Army => false,</pre>

Covered by tests:

- [omdurman-rules::src::unit_profiles::command_owns_unit_follows_tribes_and_brigades](#)
- [omdurman-net::src::lib::start_game_commands_round_trip](#)
- [omdurman-net::src::lib::legacy_start_game_without_commands_still_loads](#)
- [omdurman-net::src::lib::brigade_scope_display_and_membership](#)
- [omdurman-app::src::tests::replayed_start_game_stages_commands_and_resets_ready](#)

§1.2 – Game Scale

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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

§2 – Game Components

§2 – Game Components

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Game Components

§2.1 – The Game Maps

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The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

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Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

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The Units

File	Symbol	Code Snippet
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omdurman-types/src/lib.rs:1168 GH:omdurman-types/src/lib.rs:1168	BrigadeId	<pre> 1166 /// same field for uniform handling. 1167 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1168 pub struct BrigadeId { 1169 pub number: u8, 1170 pub nationality: BrigadeNationality, </pre>
omdurman-rules/src/unit.rs:514 GH:omdurman-rules/src/unit.rs:514	UnitProfile	<pre> 512 /// print no melee value). 513 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 514 pub struct UnitProfile { 515 pub kind: UnitKind, 516 pub identity: UnitIdentity, </pre>
omdurman-rules/src/lib.rs:14 GH:omdurman-rules/src/lib.rs:14	BrigadeId	<pre> 12 13 use omdurman_types::{ 14 BrigadeId, BrigadeNationality, DayNight, DervishTribe, HexCoord, Player, UnitKind, 15 }; 16 </pre>
omdurman-types/src/lib.rs:795 GH:omdurman-types/src/lib.rs:795	SpriteAnnotation	<pre> 793 /// as an optional overlay over the compiled 'sprite_data' fallback. 794 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 795 pub struct SpriteAnnotation { 796 pub color: SpriteColor, 797 pub faction: Option<Faction>, </pre>

Covered by tests:

- [omdurman-rules::src::unit_profiles::british_army_row_zero_specials_classify_by_counter](#)
- [omdurman-rules::src::unit_profiles::egyptian_army_row_zero_specials_classify_by_counter](#)
- [omdurman-rules::src::unit_profiles::tribe_stats_come_from_annotation](#)
- [omdurman-rules::src::unit_profiles::section_owner_dervish_sections](#)
- [omdurman-rules::src::unit_profiles::section_owner_green_sections_are_dervish](#)

§2.4 – Game Parts Inventory

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Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
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omdurman-rules/src/unit.rs:187 GH:omdurman-rules/src/unit.rs:187	WeaponClass	<pre> 185 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 186)] 187 pub enum WeaponClass { 188 /// Dervish spears and swords -- no ranged fire at all. 189 Melee,</pre>
omdurman-rules/src/unit_profiles.rs:585 GH:omdurman-rules/src/unit_profiles.rs:585	dervish_tribe	<pre> 583 /// Resolve a Dervish tribal foot counter (§2.31): Jehadia, Danagla and 584 /// Isa Zachneih fire on the rifles line; every other tribe is spear-armed. 585 pub fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> { 586 // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the 587 // Isa Zachneih unit. All other Dervish units (including leaders) are armed</pre>
omdurman-rules/src/unit_profiles.rs:350 GH:omdurman-rules/src/unit_profiles.rs:350	khalifa_abdullah	<pre> 348 /// battle (§9.322). All three are interchangeable, so they share the 349 /// `DervishArtillery` identity. 350 pub fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> { 351 let artillery = { 352 Some(Classification {</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::dervish_weapon_class_follows_the_rifles_line](#)

§2.32 – Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:126 GH:omdurman-rules/src/unit.rs:126	GunboatId	<pre> 124 /// fire; "old" gunboats do not (rulebook §2.32). 125 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 126 pub enum GunboatId { 127 /// One of the five new-type named gunboats with howitzer capability. 128 Named(NamedGunboat),</pre>
omdurman-rules/src/unit.rs:157 GH:omdurman-rules/src/unit.rs:157	NamedGunboat	<pre> 155 /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 156 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 157 pub enum NamedGunboat { 158 Sultan, 159 Melik,</pre>

omdurman-rules/src/unit.rs:170 GH:omdurman-rules/src/unit.rs:170	OldGunboat	<pre>168 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 169 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 170 pub enum OldGunboat { 171 #[strum(serialize = "Lord Kitchener")] 172 LordKitchener,</pre>
omdurman-rules/src/unit.rs:170 GH:omdurman-rules/src/unit.rs:170	GunboatId::Old	<pre>168 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 169 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 170 pub enum OldGunboat { 171 #[strum(serialize = "Lord Kitchener")] 172 LordKitchener,</pre>
omdurman-rules/src/unit.rs:132 GH:omdurman-rules/src/unit.rs:132	GunboatId::Dervish-Gunboat	<pre>130 Old(OldGunboat), 131 /// A Dervish gunboat (\$9.111, \$10.14). 132 DervishGunboat(u8), 133 } 134 </pre>

Covered

by

tests:

omdurman-rules::src::effects::tests::old_gunboat_lacks_howitzer

omdurman-rules::src::effects::tests::old_gunboat_rejected_from_howitzer_subphase

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| §3 – Getting Started

§3 – Getting Started

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Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

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§4 – Turn Sequence

§4 – Turn Sequence

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Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:191 GH:omdurman-rules/src/scalars.rs:191	GameTurnIndex	<pre>189 /// One-based Game Turn index (1, 2, ... up to the 190 scenario length) (rulebook §4). 191 #[derive(Serialize, Deserialize, Clone, Copy, 192 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 193 pub struct GameTurnIndex(pub(crate) u8); 194 195 impl GameTurnIndex {</pre>
omdurman-rules/src/turn.rs:16 GH:omdurman-rules/src/turn.rs:16	Phase	<pre>14 /// etc. 15 #[derive(Serialize, Deserialize, Clone, Copy, 16 PartialEq, Eq, Hash, Debug, Default)] 17 pub enum Phase { 18 /// Pre-game deployment (§9.2/§9.3/§10): fixed 19 /// units are placed, each side 20 /// deploys its order of battle within its legal 21 /// zone, and river</pre>
omdurman-rules/src/effects/state.rs:24 GH:omdurman-rules/src/effects/state.rs:24	GameState	<pre>22 /// All mutable state of a game in progress (rulebook 23 §4). 24 #[derive(Serialize, Deserialize, Clone, Debug)] 25 pub struct GameState { 26 pub scenario: Scenario, 27 pub current_turn: GameTurnIndex,</pre>
omdurman-rules/src/effects/state.rs:195 GH:omdurman-rules/src/effects/state.rs:195	GameState::new	<pre>193 impl GameState { 194 /// Create a fresh game state for a given scenario 195 /// (rulebook §4). 196 pub fn new(scenario: Scenario) -> Self { 197 let first = scenario_turn(scenario, 198 GameTurnIndex::new(1)); 199 // First player to *move* per scenario: 200 Campaign -- Anglo-Egyptian moves</pre>
omdurman-rules/src/effects/effect.rs:17 GH:omdurman-rules/src/effects/effect.rs:17	AdvancePhase	<pre>15 /// At end-of-turn, disrupted units recover and 16 /// per-turn tracking is 17 /// cleared. 18 AdvancePhase, 19 // -- Movement 20 -----</pre>
omdurman-rules/src/effects/dispatch.rs:177 GH:omdurman-	advance_phase	<pre>175 176 /// Advance the game state to the next phase (rulebook 177 §4).</pre>

rules/src/effects/ dispatch.rs:177		<pre>177 pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> { 178 let old_phase = state.phase; 179 #[cfg(not(feature = "kani"))]</pre>
omdurman- rules/src/effects/ dispatch.rs:346 GH:omdurman- rules/src/effects/ dispatch.rs:346	end_player_turn	<pre>344 345 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 346 pub fn end_player_turn(state: &mut GameState) { 347 #[cfg(not(feature = "kani"))] 348 debug!({</pre>
omdurman- rules/src/ scalars.rs:184 GH:omdurman- rules/src/ scalars.rs:184	GameTurnIn- dex::value	<pre>182 Self(value) 183 } 184 pub fn value(self) -> i32 { 185 self.0 186 }</pre>
omdurman- rules/src/effects/ state.rs:187 GH:omdurman- rules/src/effects/ state.rs:187	PendingMelee	<pre>185 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 186 #[derive(Serialize, Deserialize, Clone, Debug)] 187 pub struct PendingMelee { 188 pub attack: MeleeAttack, 189 pub attacker_roll: DieRoll,</pre>

Covered by tests: omdurman-rules::src::effects::tests::both_ready_auto_advances_out_of_setup
omdurman-rules::src::effects::tests::fire_combat_wrong_phase_rejected omdurman-rules::src::effects::tests::new_game_starts_in_setup
omdurman-rules::src::turn_track::scenario_turn_dispatches_correctly
omdurman-rules::src::effects::tests::turn_advances_through_phases
omdurman-rules::src::unit_profiles::game_turn_marker_cell_returns_none
omdurman-rules::src::effects::tests::declared_melee_blocks_phase_advance

18/23 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

descriptive

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Movement Phase

§5.1 – General Rules

descriptive

manual page unknown

General Rules

§5.2 – Movement Restrictions

descriptive

manual page unknown

Movement Restrictions

§5.3 – Constructing the Zariba

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Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:548 GH:omdurman-rules/src/unit.rs:548	constructing_zariba	<pre>546 /// Set while the unit is building Zariba hexsides -- -- neither offensive 547 /// fire nor melee allowed that turn (§5.3). 548 pub constructing_zariba: bool, 549 /// Set when the Royal Engineers are committed to a demolition this turn 550 /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre>
omdurman-rules/src/effects/effect.rs:153 GH:omdurman-	ConstructZariba	<pre>151 /// `hexside`, build every printed Zariba hexside they stand beside at the 152 /// end of the Anglo-Egyptian player turn. 153 ConstructZariba {</pre>

rules/src/effects/ effect.rs:153		154 unit_ids: Vec<UnitId>, 155 hexside: HexsideRef,
omdurman-rules/src/effects/dispatch.rs:43 GH:omdurman-rules/src/effects/dispatch.rs:43	apply_construct_zariba	41 } 42 GameEffect::ConstructZariba { unit_ids, hexside } => { 43 apply_construct_zariba(state, unit_ids, *hexside) 44 } 45 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target),

Covered by tests: [omdurman-rules::src::effects::tests::construct_zariba_marks_builders_and_records_hexside](#)
[omdurman-rules::src::effects::tests::zariba_builders_may_not_fire_offensively_or_melee_until_turn_end](#)
[omdurman-rules::src::effects::tests::zariba_construction_is_a_campaign_movement_phase_action](#)
[omdurman-rules::src::effects::tests::world_lens_composes_breaches_and_constructed_zariba](#)

§5.4 – Zones of Control

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Zones of Control

§5.5 – Stacking

[descriptive](#)

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Stacking

§5.11 – Movement allowances printed on units

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:63 GH:omdurman-rules/src/scalars.rs:63	MovementAllowance	61 // is a named variant. 62 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 63 pub enum MovementAllowance { 64 // Immobile (forts, wrecked gunboats). 65 Immobile = 0,
omdurman-rules/src/unit.rs:525 GH:omdurman-rules/src/unit.rs:525	UnitMovement	523 // Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 524 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 525 pub enum UnitMovement { 526 Land(MovementAllowance), 527 Gunboat(GunboatMovement),
omdurman-rules/src/scalars.rs:105 GH:omdurman-	MovementPoints	103 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, Default, 104)] 105 pub struct MovementPoints(pub(crate) i16);

rules/src/ scalars.rs:105		106 107 <code>impl std::fmt::Display for MovementPoints {</code>
omdurman- rules/src/terrain_chart.rs:21 GH:omdurman- rules/src/ terrain_chart.rs:21	terrain_effects_chart	19 <code>///</code> 20 <code>/// Source: printed Terrain Effects Chart on the</code> <code>mapsheet.</code> 21 <code>pub fn terrain_effects_chart(terrain: Terrain) -></code> <code>TerrainEntry {</code> 22 <code> match terrain {</code> 23 <code> Terrain::Clear { .. } => TerrainEntry {</code>
omdurman- rules/src/terrain_chart.rs:77 GH:omdurman- rules/src/ terrain_chart.rs:77	movement_cost	75 <code>/// Convenience: get the movement cost for a terrain</code> <code>type (rulebook §5.11, Terrain Effects Chart).</code> 76 <code>/// Returns `None` for impassable terrain (Nile).</code> 77 <code>pub fn movement_cost(terrain: Terrain) -></code> <code>Option<MovementAllowance> {</code> 78 <code> terrain_effects_chart(terrain).movement_cost</code> 79 <code>}</code>
omdurman- rules/src/terrain_chart.rs:87 GH:omdurman- rules/src/ terrain_chart.rs:87	movement_cost_with_ro	85 <code>/// movement overlay only -- combat/LOS use the</code> <code>underlying terrain ("Per</code> 86 <code>/// other terrain in hex"). The Nile stays impassable</code> <code>to land units.</code> 87 <code>pub fn movement_cost_with_road(terrain: Terrain,</code> <code>along_road: bool) -> Option<MovementAllowance> {</code> 88 <code> if along_road && !terrain.is_nile() {</code> 89 <code> Some(MovementAllowance::One)</code>
omdurman- types/src/ lib.rs:583 GH:omdurman- types/src/lib.rs:583	Terrain::has_road	581 582 <code>/// Whether this hex has any road touching it.</code> 583 <code>pub fn has_road(self) -> bool {</code> 584 <code> !matches!(self.road(), Road::None)</code> 585 <code>}</code>
omdurman- types/src/ lib.rs:520 GH:omdurman- types/src/lib.rs:520	Terrain::passable_by_land	518 519 <code>/// Whether this terrain may be entered by land</code> <code>units (rulebook §5.11).</code> 520 <code>pub fn passable_by_land(self) -> bool {</code> 521 <code> !self.is_nile()</code> 522 <code>}</code>
omdurman- types/src/ lib.rs:588 GH:omdurman- types/src/lib.rs:588	Terrain::is_crossroad	586 587 <code>/// Whether roads converge at this hex's centre.</code> 588 <code>pub fn is_crossroad(self) -> bool {</code> 589 <code> matches!(self.road(), Road::Crossroad)</code> 590 <code>}</code>
omdurman- rules/src/ board.rs:462 GH:omdurman- rules/src/ board.rs:462	BoardInfo::roads	460 <code>tiles: tiles.into_iter().collect(),</code> 461 <code>hexsides: Vec::new(),</code> 462 <code>roads: Vec::new(),</code> 463 <code>excluded: BTreeSet::new(),</code> 464 <code>overlay: default_overlay(),</code>

Proven by: `omdurman-rules::src::terrain_chart::movement_column_matches_the_printed_chart`
`omdurman-types::src::lib::terrain_road_state_round_trips`

Covered by tests: `omdurman-rules::src::effects::tests::road_rate_applies_only_along_the_road`
`omdurman-rules::src::effects::tests::crossing_a_hexside_costs_the_chart_surcharge`
`omdurman-rules::src::terrain_chart::step_cost_adds_the_crossed_hexside`
`omdurman-rules::src::terrain_chart::engine_matches_the_transcribed_terrain_effects_chart`
`omdurman-rules::src::effects::tests::extreme_coordinates_are_rejected_without_panicking`
`omdurman-rules::src::effects::tests::move_cost_is_engine_computed_and_caller_cost_ignored`
`omdurman-rules::src::effects::tests::move_path_must_be_contiguous_and_end_at_destination`
`omdurman-rules::src::terrain_chart::clear_terrain_no_bonus`
`omdurman-rules::src::terrain_chart::nile_is_impassable` `omdurman-rules::src::terrain_chart::rough_movement_and_defense`
`omdurman-rules::src::terrain_chart::swamp_movement_and_defense`
`omdurman-rules::src::terrain_chart::hilltop_movement_and_defense`
`omdurman-rules::src::terrain_chart::huts_movement_and_defense`
`omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart`

omdurman-rules::src::terrain_chart::movement_cost_with_road_always_one
 omdurman-rules::src::effects::tests::land_unit_may_not_enter_nile
 omdurman-rules::src::terrain_chart::movement_cost_without_road_matches_terrain
 omdurman-rules::src::effects::tests::movement_cost_for_uses_terrain
 omdurman-rules::src::effects::tests::movement_cost_for_road_costs_one
 omdurman-rules::src::terrain_chart::road_gives_crossroad omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds
 omdurman-rules::src::terrain_chart::terrain_chart_road_always_costs_one
 omdurman-rules::src::terrain_chart::trees_cost_one_and_give_no_defense

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

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A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:25 GH:omdurman-rules/src/effects/state/movement.rs:25	can_move_unit	<pre> 23 /// Returns the same `RuleError` the `MoveUnit` effect would on rejection. 24 /// Lets the UI gate input without mutating or duplicating the rules. 25 pub fn can_move_unit(&self, unit_id: UnitId, cost: MovementPoints) -> Result<(), RuleError> { 26 let unit = self.unit_or_err(unit_id)?; 27 self.movement_preconditions(unit)?; </pre>
omdurman-rules/src/effects/state.rs:393 GH:omdurman-rules/src/effects/state.rs:393	mp_spent	<pre> 391 392 /// Movement points `unit_id` has already spent this turn (§5.11/§5.12). 393 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { 394 395 self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0) </pre>

Covered by tests: omdurman-rules::src::effects::tests::cumulative_move_cost_may_not_exceed_allowance
 omdurman-rules::src::effects::tests::wrong_faction_move_is_rejected
 omdurman-rules::src::effects::tests::wrong_faction_gunboat_move_is_rejected

§5.13 – No MP accumulation between turns

implemented

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A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:346 GH:omdurman-rules/src/effects/dispatch.rs:346	end_player_turn	<pre> 344 345 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 346 pub fn end_player_turn(state: &mut GameState) { 347 #[cfg(not(feature = "kani"))] 348 debug!(</pre>

Covered by tests: omdurman-rules::src::effects::tests::disrupted_units_recover_at_the_end_of_their_owners_turn
 omdurman-rules::src::effects::tests::unused_movement_points_do_not_carry_over

§5.21 – Friendlies transport via gunboat

implemented

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In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:339 GH:omdurman-rules/src/unit.rs:339	is_friendlys	<pre>337 /// "Friendlies" units obey several special rules 338 (\$5.21, \$5.23, \$6.52, 339 /// \$9.14 victory conditions). 340 pub fn is_friendlys(&self) -> bool { 341 matches!(342 self,</pre>
omdurman-rules/src/effects/state/movement.rs:689 GH:omdurman-rules/src/effects/state/movement.rs:689	friendlys_transport_offers	<pre>687 /// [<code>GameEffect::FriendlysTransport</code>] so the UI 688 offers exactly what the 689 /// engine accepts. 690 pub fn friendlys_transport_offers(&self, selected: 691 Option<UnitId>) -> Vec<FriendlysAction> { 692 let candidates: Vec<FriendlysAction> = match 693 self.friendlys_transport { 694 None => {</pre>
omdurman-rules/src/unit.rs:545 GH:omdurman-rules/src/unit.rs:545	loaded_on	<pre>543 pub disrupted: bool, 544 /// `Some(gunboat)` after a "Friendlies" unit loads 545 onto a gunboat (§5.21). 546 pub loaded_on: Option<UnitId>, 547 /// Set while the unit is building Zariba hexsides 548 -- neither offensive 549 /// fire nor melee allowed that turn (§5.3).</pre>
omdurman-rules/src/effects/effect.rs:180 GH:omdurman-rules/src/effects/effect.rs:180	FriendlysTransport	<pre>178 179 /// Load/disembark the "Friendlies" brigade via 180 gunboat (rulebook §5.21). 181 FriendlysTransport(crate::FriendlysAction), 182 // -- Optional rules 183 -----</pre>
omdurman-rules/src/effects/river.rs:10 GH:omdurman-rules/src/effects/river.rs:10	apply_friendlys_transport	<pre>8 /// * `Disembark`: the unit lands on the chosen west- 9 bank hex, paying its 10 /// terrain cost like any first hex entered, and the 11 mission ends. 12 pub fn apply_friendlys_transport(13 state: &mut GameState, 14 action: FriendlysAction,</pre>
omdurman-rules/src/transport.rs:17 GH:omdurman-rules/src/transport.rs:17	FriendlysAction	<pre>15 /// two, the gunboat carries the unit wherever it 16 moves. 17 #[derive(Serialize, Deserialize, Clone, Copy, 18 PartialEq, Eq, Debug)] 19 pub enum FriendlysAction { 20 /// Turn N: "a 'Friendlies' unit and any Anglo- 21 Egyptian gunboat start 22 /// their turn adjacent" -- the unit loads onto 23 (stacks with) the gunboat.</pre>

omdurman-rules/src/transport.rs:33 GH:omdurman-rules/src/transport.rs:33	TransportState	<pre> 31 /// A Friendlies transport under way (§5.21), stored on GameState`. 32   #[derive(Serialize, Deserialize, Clone, Copy,      PartialEq, Eq, Debug)] 33   pub enum TransportState { 34       /// `unit` has been aboard `gunboat` since turn      `since` (the load turn, 35 /// N): the gunboat may carry it on turn N+1, and it may disembark from </pre>
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Covered by tests: [omdurman-rules::src::effects::tests::a_mined_gunboat_scores_and_takes_its_friendlies_down](#)
[omdurman-rules::src::effects::tests::friendlies_transport_offer_load_requires_prerequisites](#)
[omdurman-rules::src::effects::tests::friendlies_cross_the_nile_over_three_turns](#)

§5.22 – Land units may never enter a Nile River hex

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With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/ef-fects/state/movement.rs:43 GH:omdurman-rules/src/ef-fects/state/movement.rs:43	can_move_unit_to	<pre> 41 /// 42 /// Stacking (§5.51) is checked when the move is applied, not here. 43 pub fn can_move_unit_to(44 &self, 45 unit_id: UnitId, </pre>
omdurman-rules/src/ef-fects/state/setup.rs:239 GH:omdurman-rules/src/effects/state/setup.rs:239	in_deployment_zone	<pre> 237 /// areas ([`Self::campaign_set_up_area`]); nothing for the 238 /// Anglo-Egyptians. 239 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 240 // No board attached -> permissive (unit tests, unbound session). 241 if self.board.terrain.is_empty() { </pre>

Covered by tests: [omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile](#)
[omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine](#)
[omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_nile](#)
[omdurman-rules::src::effects::tests::a_gunboat_is_not_offered_a_land_advance](#)

§5.23 – Walled city entry restrictions

implemented

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Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
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omdurman-types/src/lib.rs:223 GH:omdurman-types/src/lib.rs:223	HexsideRef	<pre> 221 /// data by [`HexsideRef`]. 222 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 223 pub struct HexsideRef { 224 pub a: HexCoord, 225 pub b: HexCoord, </pre>
omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	HexsideKind	<pre> 256 strum::EnumIter, 257)] 258 pub enum HexsideKind { 259 /// City wall (Khartoum, walled city of Omdurman). Blocks LOS, blocks 260 /// movement except across gates/breaches (§5.23), blocks ZOC into the city </pre>
omdurman-types/src/lib.rs:336 GH:omdurman-types/src/lib.rs:336	blocks_movement	<pre> 334 /// `omdurman-rules`. The trench *end* variants are therefore intentionally 335 /// not blocking. 336 pub fn blocks_movement(self) -> bool { 337 matches!(338 self, </pre>
omdurman-rules/src/unit.rs:365 GH:omdurman-rules/src/unit.rs:365	may_enter_walled_city	<pre> 363 /// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the 364 /// walled city *except* gunboats and "Friendlies". 365 pub fn may_enter_walled_city(&self) -> bool { 366 match self { 367 // §5.23 Dervish: Khalifa, artillery, Taiasha. </pre>
omdurman-rules/src/board.rs:369 GH:omdurman-rules/src/board.rs:369	is_walled_city	<pre> 367 /// are its seeds). The set is derived once from the board data, replacing 368 /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. 369 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 370 // Membership in the precomputed enclosed area (see `walled_city`). 371 // Palace/Tomb hexes are always part of it (the they are the seeds). </pre>
omdurman-rules/src/effects/error.rs:116 GH:omdurman-rules/src/effects/error.rs:116	WalledCityEntry	<pre> 114 115 #[error("unit {0} is not eligible to enter the walled city of Omdurman at {1} (§5.23)")] 116 WalledCityEntry(UnitId, HexCoord), 117 118 #[error("movement cost {cost} exceeds allowance {allowance}")] </pre>

Proven

by:

[omdurman-types::src::lib::hexside_ref_is_order_independent](#) [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)

Covered by tests:

[omdurman-rules::tests::walled_city_entry::only_the_khalifas_men_pass_the_gates_of_omdurman](#)
[omdurman-rules::src::effects::tests::a_wall_blocks_every_step_not_just_the_endpoints](#)
[omdurman-rules::src::effects::tests::can_move_rejects_wall_hexside](#) [omdurman-rules::src::effects::tests::can_move_allows_gate_hexside](#)
[omdurman-rules::src::effects::tests::walled_city_entry_allows_khalifa](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_unauthorized_dervish](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_ae_gunboat](#)
[omdurman-rules::src::effects::tests::walled_city_entry_not_enforced_for_fok](#)
[omdurman-rules::src::board_data::campaign_walled_city_is_enclosed_by_walls](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)
[omdurman-rules::src::effects::tests::a_retreat_needs_an_open_path_and_may_join_friends](#)

§5.24 – Gunboat upstream/downstream movement

implemented

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Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:263 GH:omdurman-rules/src/unit.rs:263	GunboatMovement	<pre> 261 /// the turn. 262 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 263 pub struct GunboatMovement { 264 pub upstream: MovementAllowance, 265 pub downstream: MovementAllowance, </pre>
omdurman-types/src/lib.rs:1092 GH:omdurman-types/src/lib.rs:1092	is_boat	<pre> 1090 1091 /// Gunboats use the split upstream/downstream movement allowance (§5.24). 1092 pub fn is_boat(self) -> bool { 1093 matches!(self, UnitKind::Gunboat { .. }) 1094 } </pre>

Proven by: [omdurman-types::src::lib::unit_kind_fire_and_movement_capability Law is exact](#)
[omdurman-types::src::lib::hex_direction_from_index is total mod six](#)
[omdurman-types::src::lib::nile_flow_rotation_composes_and_never_leaves_the_nile](#)

Covered by tests: [omdurman-rules::src::unit_profiles::boat_annotation_yields_split_gunboat_movement](#)
[omdurman-rules::src::effects::tests::gunboat_upstream_cap_is_sticky_across_moves](#)
[omdurman-app::src::picker::movement::gunboat_cap_gate_mirrors_the_engine](#)
[omdurman-app::src::picker::movement::remaining_mp_respects_the_sticky_upstream_cap](#)
[omdurman-rules::src::board::step_direction_oblique_steps_follow_the_current](#)

§5.25 – Dervish forts may not move

implemented

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Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:529 GH:omdurman-rules/src/unit.rs:529	Immobile	<pre> 527 Gunboat(GunboatMovement), 528 /// Forts may not move once placed (§5.25). 529 Immobile, 530 } 531 </pre>
omdurman-types/src/lib.rs:1049 GH:omdurman-types/src/lib.rs:1049	UnitKind::Fort	<pre> 1047 /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 1048 /// May not move once placed (§5.25). 1049 Fort { fire: i32, melee: i32 }, 1050 /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 1051 DervishLeader { </pre>

omdurman-rules/src/unit.rs:529 GH:omdurman-rules/src/unit.rs:529	UnitMovement::Immobile	<pre> 527 Gunboat(GunboatMovement), 528 /// Forts may not move once placed (§5.25). 529 Immobile, 530 } 531 </pre>
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Covered by tests: [omdurman-rules::src::effects::tests::forts_are_never_advance_eligible](#)
[omdurman-rules::src::effects::tests::immobile_fort_rejects_move_unit](#)

§5.26 – Units stop on entering enemy ZOC

implemented

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Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:43 GH:omdurman-rules/src/effects/state/movement.rs:43	can_move_unit_to	<pre> 41 /// 42 /// Stacking (§5.51) is checked when the move is applied, not here. 43 pub fn can_move_unit_to(44 &self, 45 unit_id: UnitId, </pre>
omdurman-rules/src/effects/state/stacking.rs:222 GH:omdurman-rules/src/effects/state/stacking.rs:222	hex_in_enemy_zoc	<pre> 220 /// does not extend into or out of a Nile hex. With no board loaded these 221 /// reduce to the plain adjacency rule. 222 pub fn hex_in_enemy_zoc(223 &self, 224 hex: HexCoord, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)
[omdurman-rules::src::effects::tests::zoc_transit_check_uses_the_actual_path](#)
[omdurman-rules::src::effects::tests::entering_reinforcements_stop_in_zoc_and_overrun_lone_leaders](#)

§5.41 – All units except AE leaders exert ZOC

implemented

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All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/stacking.rs:114 GH:omdurman-rules/src/effects/state/stacking.rs:114	unit_projects_zoc_rule	<pre> 112 /// * A fort projects ZOC out of its hex even when unoccupied (§5.44), 113 /// modelled by the fort unit projecting normally. 114 pub fn unit_projects_zoc_rule(115 unit: &UnitPlacement, 116 mover_player: Player, </pre>
omdurman-rules/src/	ZocReason	<pre> 582 /// Used by the engine when answering "is this hex in an enemy ZOC?". </pre>

unit.rs:584 GH:omdurman-rules/src/unit.rs:584		<pre> 583 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 584 pub enum ZocReason { 585 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 586 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-rules/src/effects/state/stacking.rs:204 GH:omdurman-rules/src/effects/state/stacking.rs:204	unit_projects_zoc	<pre> 202 /// §5.44) need the game map, which the engine does not hold; the app layers 203 /// those on top. This is the position/kind/ disruption core of the rule. 204 pub fn unit_projects_zoc(205 &self, 206 unit: &UnitPlacement, </pre>
omdurman-rules/src/effects/state/stacking.rs:296 GH:omdurman-rules/src/effects/state/stacking.rs:296	zoc_hexes	<pre> 294 /// ZOC covers a given hex; this function returns *which* hexes a 295 /// specific unit covers. 296 pub fn zoc_hexes(297 &self, 298 unit: &UnitPlacement, </pre>

Proven by: [omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses](#)
[omdurman-types::src::lib::player_opponent_is_a_fixed_point_free_involution](#)

Covered by tests: [omdurman-rules::src::effects::tests::zoc_hexes_empty_for_anglo_egyptian_leader](#)
[omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions](#)
[omdurman-rules::src::effects::tests::zoc_hexes_empty_for_disrupted_unit](#)
[omdurman-rules::src::effects::tests::zoc_hexes_matches_hex_in_enemy_zoc](#) [omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile](#)
[omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor](#)

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

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There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:544 GH:omdurman-rules/src/effects/state/movement.rs:544	movement_cost_for	<pre> 542 /// 543 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 544 pub fn movement_cost_for(545 &self, 546 unit: &UnitPlacement, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::entering_enemy_zoc_costs_no_extra_mp](#)

§5.43 – Units stop when entering enemy ZOC

implemented

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/ef-fects/state/movement.rs:43 GH:omdurman-rules/src/ef-fects/state/movement.rs:43	can_move_unit_to	<pre> 41 /// 42 /// Stacking (§5.51) is checked when the move is applied, not here. 43 pub fn can_move_unit_to(44 &self, 45 unit_id: UnitId, </pre>
omdurman-rules/src/ef-fects/state/stacking.rs:222 GH:omdurman-rules/src/ef-fects/state/stacking.rs:222	hex_in_enemy_zoc	<pre> 220 /// does not extend into or out of a Nile hex. With no board loaded these 221 /// reduce to the plain adjacency rule. 222 pub fn hex_in_enemy_zoc(223 &self, 224 hex: HexCoord, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

manual page unknown

ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:584 GH:omdurman-rules/src/unit.rs:584	ZocReason	<pre> 582 /// Used by the engine when answering "is this hex in an enemy ZOC?". 583 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 584 pub enum ZocReason { 585 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 586 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-types/src/lib.rs:263 GH:omdurman-types/src/lib.rs:263	Wall	<pre> 261 /// (§5.44), blocks melee (§7.2), blocks advance- after-combat (§6.82). 262 #[default] 263 Wall, 264 /// Gate hexside in a wall. ZOC extends *out of* the walled city through 265 /// gates but not into it (§5.44). Melee may be made through a gate (§7.2). </pre>
omdurman-types/src/lib.rs:272	Khor	<pre> 270 /// Khor -- gully/wadi. ZOCs do not extend across (§5.44); advance after 271 /// combat may not cross (§6.82). 272 Khor, </pre>

GH:omdurman-types/src/lib.rs:272		<pre> 273 /// Crest line. Blocks LOS unless the firer is on the higher side 274 /// (§6.3 note 7). </pre>
omdurman-rules/src/effects/state/stacking.rs:247 GH:omdurman-rules/src/effects/state/stacking.rs:247	zoc_extends	<pre> 245 /// only outward ("out of, but not into, a walled city hex"), across a 246 /// breach both ways. 247 pub fn zoc_extends(&self, unit: &UnitPlacement, into: HexCoord) -> bool { 248 use omdurman_types::HexsideKind; 249 let from = unit.position; </pre>
omdurman-rules/src/effects/state/stacking.rs:204 GH:omdurman-rules/src/effects/state/stacking.rs:204	unit_projects_zoc	<pre> 202 /// §5.44) need the game map, which the engine does not hold; the app layers 203 /// those on top. This is the position/kind/ disruption core of the rule. 204 pub fn unit_projects_zoc(205 &self, 206 unit: &UnitPlacement, </pre>
omdurman-types/src/lib.rs:354 GH:omdurman-types/src/lib.rs:354	HexsideKind::blocks_zoc	<pre> 352 /// lets a Zariba unit's ZOC out). This predicate captures the "does not 353 /// extend across" default. 354 pub fn blocks_zoc(self) -> bool { 355 matches!(356 self, </pre>
omdurman-rules/src/effects/state/stacking.rs:222 GH:omdurman-rules/src/effects/state/stacking.rs:222	hex_in_enemy_zoc	<pre> 220 /// does not extend into or out of a Nile hex. With no board loaded these 221 /// reduce to the plain adjacency rule. 222 pub fn hex_in_enemy_zoc(223 &self, 224 hex: HexCoord, </pre>
omdurman-rules/src/effects/state/stacking.rs:247 GH:omdurman-rules/src/effects/state/stacking.rs:247	zoc_extends	<pre> 245 /// only outward ("out of, but not into, a walled city hex"), across a 246 /// breach both ways. 247 pub fn zoc_extends(&self, unit: &UnitPlacement, into: HexCoord) -> bool { 248 use omdurman_types::HexsideKind; 249 let from = unit.position; </pre>

Proven by: `omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses`
`omdurman-types::src::lib::hexside_blocking_classifiers_are_exact`

Covered by tests: `omdurman-rules::src::effects::tests::zoc_extends_out_of_but_not_into_the_zariba`
`omdurman-rules::src::effects::tests::zoc_extends_out_of_but_not_into_buildings_and_forts`
`omdurman-rules::src::effects::tests::zoc_crosses_walls_and_gates_only_outward`
`omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile` `omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor`
`omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions`

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

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No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:603 GH:omdurman-rules/src/unit.rs:603	OverLimit	<pre> 601 /// (§6.51: a Dervish unit occupying his hex eliminates him). 602 #[error("hex stack exceeds the four-unit limit [\$5.51]")] 603 OverLimit, 604 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 605 #[error("gunboats may not stack with non-gunboat units [\$5.51]")] </pre>
omdurman-rules/src/unit.rs:606 GH:omdurman-rules/src/unit.rs:606	GunboatStack	<pre> 604 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 605 #[error("gunboats may not stack with non-gunboat units [\$5.51]")] 606 GunboatStack, 607 /// "Units of different Dervish tribes may not stack together" (§5.52). 608 /// The Dervish artillery (§9.322's three guns) is not a tribe but is not </pre>
omdurman-rules/src/unit.rs:616 GH:omdurman-rules/src/unit.rs:616	EnemyCohabitation	<pre> 614 /// lone Anglo-Egyptian leader is exempt (§6.51). #[error("enemy units may not share a hex; melee, not movement, engages them [\$5.51, §7.1]")] 616 EnemyCohabitation, 617 /// "If Dervish leaders elect to stack, they may only stack with units of 618 /// their command (i.e. colour)" (§5.53). </pre>
omdurman-rules/src/ef-fects/state/stacking.rs:155 GH:omdurman-rules/src/ef-fects/state/stacking.rs:155	check_stacking	<pre> 153 /// Dervish artillery is its own group) may not stack together. 154 /// * §5.53 -- a Dervish leader may stack only with units of its command. 155 pub fn check_stacking(156 &self, 157 mover: &UnitPlacement, </pre>
omdurman-rules/src/ef-fects/state/stacking.rs:21 GH:omdurman-rules/src/ef-fects/state/stacking.rs:21	stacking_rule	<pre> 19 /// artillery as its own group) may not share a hex. 20 /// * §5.53: a Dervish leader stacks only with units of its command. 21 pub fn stacking_rule(occupants: &[&UnitPlacement]) -> Result<(), crate::StackingError> { 22 use crate::StackingError; 23 </pre>

Proven by: [omdurman-rules::src::effects::stacking_rule_cohabitation_is_exact](#)
[omdurman-rules::src::effects::stacking_rule_is_symmetric](#) [omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking](#)
[omdurman-rules::src::effects::stacking_rule_limit_is_four_counted_units](#)
[omdurman-rules::src::effects::stacking_rule_gunboat_never_shares](#)

Covered by tests: [omdurman-app::src::melee::one_click_advances_the_stack_up_to_the_stacking_limit](#)
[omdurman-rules::src::effects::tests::stacking_over_limit_rejected](#)
[omdurman-rules::src::effects::tests::mid_move_stacking_allows_pass_through](#)
[omdurman-rules::src::effects::tests::mid_move_stacking_rejects_over_limit_destination](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_stacking_violation](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_allows_leaders_stacking](#)
[omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_enemy_cohabitation](#)
[omdurman-rules::src::effects::tests::ae_garrison_stacks_freely_under_gordon](#)
[omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking](#)

§5.52 – Different Dervish tribes may not stack together

[implemented](#)

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The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:612 GH:omdurman-rules/src/unit.rs:612	DervishTribeMix	<pre>610 /// [`UnitIdentity::dervish_stacking_group`]). 611 #[error("Dervish units of different tribes may not stack [\$5.52]")] 612 DervishTribeMix, 613 /// A unit may never share a hex with enemy units (\$5.51, §7.1); only the 614 /// lone Anglo-Egyptian leader is exempt (§6.51).</pre>
omdurman-rules/src/unit.rs:418 GH:omdurman-rules/src/unit.rs:418	dervish_stacking_group	<pre>416 /// `None` for every unit the §5.52 law does not constrain: Dervish 417 /// leaders, forts and gunboats, and all Anglo-Egyptian units. 418 pub fn dervish_stacking_group(&self) -> Option<DervishStackingGroup> { 419 match self { 420 UnitIdentity::DervishTribal { tribe } => Some(DervishStackingGroup::Tribe(*tribe)),</pre>

Proven by: [omdurman-rules::src::effects::stacking_rule_group_purity_is_exact](#)

Covered by tests: [omdurman-rules::src::unit_profiles::green_sections_are_mulazmin_tribal_units](#)
[omdurman-rules::src::effects::tests::deploy_rejects_dervish_tribe_mix](#)
[omdurman-rules::src::effects::tests::deploy_rejects_hadendowa_on_dervish_gun](#)
[omdurman-rules::src::effects::tests::dervish_guns_stack_with_guns_and_their_leader](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state](#)

§5.53 – Leader stacking with command colour only

implemented

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Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:620 GH:omdurman-rules/src/unit.rs:620	DervishLeaderCommandMismatch	<pre>618 /// their command (i.e. colour)" (§5.53). 619 #[error("Dervish leader may only stack with units of their own command [\$5.53]")] 620 DervishLeaderCommandMismatch, 621 } 622 </pre>

Proven by: [omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking](#)

Covered by tests: [omdurman-rules::src::effects::tests::dervish_leader_stacks_only_with_command_colour](#)
[omdurman-rules::src::unit_profiles::command_owns_unit_follows_tribes_and_brigades](#)
[omdurman-rules::src::effects::tests::dervish_leaders_of_different_commands_do_not_stack](#)

§5.54 – Anglo-Egyptian Brigade Integrity

[implemented](#)

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:626 GH:omdurman-rules/src/unit.rs:626	BrigadeIntegrity	<pre> 624 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 625 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 626 pub enum BrigadeIntegrity { 627 None, 628 Integrated(BrigadeId), </pre>
omdurman-rules/src/unit.rs:480 GH:omdurman-rules/src/unit.rs:480	brigade_integrity	<pre> 478 /// Only a full stack of four battalions qualifies. Three or fewer may still 479 /// stack and fire, but they receive no brigade- integrity bonus. 480 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 481 // Brigades as printed (§5.54): 1E is II Egyptian plus three Sudanese 482 // battalions (see `UnitIdentity::brigade`). </pre>
omdurman-rules/src/unit.rs:626 GH:omdurman-rules/src/unit.rs:626	FireModifier::BrigadeIntegrity	<pre> 624 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 625 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 626 pub enum BrigadeIntegrity { 627 None, 628 Integrated(BrigadeId), </pre>
omdurman-rules/src/unit.rs:20 GH:omdurman-rules/src/unit.rs:20	BattalionOrdinal	<pre> 18 /// brigade integrity requires all four stacked in one hex (§5.54). 19 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 20 pub enum BattalionOrdinal { 21 First = 1, 22 Second = 2, </pre>
omdurman-types/src/lib.rs:1168 GH:omdurman-types/src/lib.rs:1168	BrigadeId	<pre> 1166 /// same field for uniform handling. 1167 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1168 pub struct BrigadeId { 1169 pub number: u8, 1170 pub nationality: BrigadeNationality, </pre>

Proven by: [omdurman-rules::src::verification::brigade_integrity_requires_all_four_distinct_battalions_of_one_brigade](#)

Covered by tests: [omdurman-rules::src::effects::tests::brigade_integrity_survives_combined_fire](#)
[omdurman-rules::src::unit_profiles::egyptian_division_brigades_follow_the_printed_designations](#)
[omdurman-rules::src::unit_profiles::brigade_designation_ignored_for_non_infantry](#)
[omdurman-rules::src::unit_profiles::printed_brigade_designation_overrides_column](#)
[omdurman-rules::src::unit_profiles::section_owner_anglo_egyptian_sections](#)
[omdurman-rules::src::tests::brigade_integrity_four_battalions_returns_integrated](#)
[omdurman-rules::src::unit_profiles::ae_infantry_fourth_battalion_from_col_3](#)
[omdurman-rules::src::tests::brigade_integrity_empty_slice](#) [omdurman-rules::src::tests::brigade_integrity_friendly_returns_none](#)
[omdurman-rules::src::tests::brigade_integrity_three_battalions_returns_none](#)
[omdurman-rules::src::tests::unit_identity_brigade_and_battalion_accessors](#)
[omdurman-rules::src::unit_profiles::ae_infantry_brigade_number_three_from_col_7](#)
[omdurman-rules::src::tests::brigade_integrity_non_infantry_returns_none](#)
[omdurman-rules::src::tests::brigade_integrity_mixed_brigades_returns_none](#)
[omdurman-rules::src::unit_profiles::ae_infantry_third_battalion_from_col_2](#)
[omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived](#)



23/31 implemented

§6 – Fire Combat Phase

§6 – Fire Combat Phase

descriptive

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Fire Combat Phase

§6.1 – General Rules

descriptive

manual page unknown

General Rules

§6.2 – How To Have Fire Combat

descriptive

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How To Have Fire Combat

§6.3 – Line of Sight Table

implemented

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Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:57 GH:omdurman-rules/src/los_table.rs:57	LosLevel	<pre>55 serde::Serialize, serde::Deserialize, Clone, Copy, 56 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57)] 58 pub enum LosLevel { 59 Ground, 59 Rough,</pre>
omdurman-rules/src/los_table.rs:81 GH:omdurman-rules/src/los_table.rs:81	LosFeature	<pre>79 /// authored RON spellings. 80 #[derive(serde::Serialize, serde::Deserialize, Clone, 81 Copy, PartialEq, Eq, Hash, Debug)] 82 pub enum LosFeature { 83 /// A hex containing units (gunboats/forts excluded 83 Units,</pre>
omdurman-rules/src/los_table.rs:117 GH:omdurman-rules/src/los_table.rs:117	LosCondition	<pre>115 /// A positional condition from the LOS table Detail 116 footnotes. 117 #[derive(serde::Serialize, serde::Deserialize, Clone, 118 Copy, PartialEq, Eq, Hash, Debug)] 119 pub enum LosCondition { 120 /// (1) Blocks only if the ray passes through more</pre>

		than two such features. <pre>119 MoreThanTwo,</pre>
omdurman-rules/src/los_table.rs:197 GH:omdurman-rules/src/los_table.rs:197	los_level	<pre>195 /// 196 /// For all other units, the level is derived from the terrain at `hex`. 197 pub fn los_level_for_unit(198 kind: UnitKind, 199 hex: HexCoord,</pre>
omdurman-rules/src/los_table.rs:197 GH:omdurman-rules/src/los_table.rs:197	los_level_for_unit	<pre>195 /// 196 /// For all other units, the level is derived from the terrain at `hex`. 197 pub fn los_level_for_unit(198 kind: UnitKind, 199 hex: HexCoord,</pre>
omdurman-rules/src/los_table.rs:235 GH:omdurman-rules/src/los_table.rs:235	blocking_rules	<pre>233 /// always blocks. Indexing is in-bounds by construction (both enums have 234 exactly three variants). 235 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 236 crate::tables_data::LOS_CELLS[firer.index()] [target.index()] 237 }</pre>
omdurman-rules/src/los_table.rs:584 GH:omdurman-rules/src/los_table.rs:584	has_los	<pre>582 /// board-only tools). 583 #[allow(clippy::too_many_arguments)] 584 pub fn has_los(585 board: &crate::board::BoardInfo, 586 from: HexCoord,</pre>
omdurman-types/src/lib.rs:301 GH:omdurman-types/src/lib.rs:301	HexsideKind::blocks_los	<pre>299 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 300 /// in `omdurman_rules::los_table`, not by this predicate. 301 pub fn blocks_los(self) -> bool { 302 matches!(self, HexsideKind::Wall HexsideKind::Crest) 303 }</pre>
omdurman-types/src/lib.rs:301 GH:omdurman-types/src/lib.rs:301	Terrain::blocks_los	<pre>299 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 300 /// in `omdurman_rules::los_table`, not by this predicate. 301 pub fn blocks_los(self) -> bool { 302 matches!(self, HexsideKind::Wall HexsideKind::Crest) 303 }</pre>
omdurman-types/src/lib.rs:537 GH:omdurman-types/src/lib.rs:537	Terrain::is_los_trees	<pre>535 /// (§6.3 note 1). Retained for compatibility; the full LOS engine 536 /// checks `Terrain::Trees` directly. 537 pub fn is_los_trees(self) -> bool { 538 matches!(self, Terrain::Trees { .. }) 539 }</pre>

Proven

by: [omdurman-types::src::lib::line_between_forms_a_connected_ray](#) [omdurman-types::src::lib::hexside_blocking_classifiers_are_exact](#)
[omdurman-rules::src::los_table::los_level_depends_only_on_the_ground](#)
[omdurman-rules::src::los_table::los_level_overrides_hold_for_gunboats_and_forts](#)
[omdurman-rules::src::los_table::blocking_grid_is_total_and_walls_block_ground_firers](#)

Covered

by

tests:

[omdurman-rules::src::los_table::los_ray_is_the_straight_line_on_the_campaign_board](#)
[omdurman-rules::src::los_table::los_level_mapping](#)
[omdurman-rules::src::los_table::blocking_rules_all_cells_covered](#) [omdurman-rules::src::los_table::has_los_empty_board_is_clear](#)
[omdurman-rules::src::los_table::has_los_adjacent_clear](#) [omdurman-rules::src::los_table::has_los_howitzer_bypasses](#)
[omdurman-rules::src::los_table::has_los_wall_hexside_blocks_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_gate_hexside_passes](#) [omdurman-rules::src::los_table::has_los_breach_hexside_passes](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)

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omdurman-rules::src::los_table::has_los_rough_intervening_blocks_ground_to_ground
omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzers_always_has_los
omdurman-rules::src::los_table::los_howitzers_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected
omdurman-rules::src::effects::tests::a_crest_halfway_along_the_line_blocks_ground_to_hilltop_fire

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§6.4 – Fire Combat Sequence

implemented

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Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

****6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):**** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

****6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):**** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-rules/src/turn.rs:44 GH:omdurman-rules/src/turn.rs:44	FireSubPhase	<pre> 42 43 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 44 pub enum FireSubPhase { 45 /// Direct fire (§6.41). Both sides participate in this sub-phase. 46 DirectFire, </pre>
omdurman-rules/src/effects/effect.rs:73 GH:omdurman-rules/src/effects/effect.rs:73	FireCombat	<pre> 71 /// - Firers marked as fired; target hex marked as fired-at. 72 /// - Victory points awarded for eliminations. 73 FireCombat { attack: FireAttack, roll: DieRoll }, 74 75 /// Resolve a howitzer bombardment (two rolls: CRT + impact scatter) </pre>

Covered by tests: [omdurman-rules::src::effects::tests::turn_advances_through_phases](#)

§6.5 – Special Unit Capabilities

descriptive
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Special Unit Capabilities

****6.51) Leader Units:**** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi’s tomb hex if it is to be taken from the Dervish player (see 9.14).

****6.52) Anglo-Egyptian “Friendlies” Brigade:**** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive
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Special Artillery Capabilities

§6.7 – Defensive Fire

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Defensive Fire

In Defensive Fire phase, all of the non-moving player’s units may fire at any of the moving player’s units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/melee.rs:177 GH:omdurman-rules/src/effects/state/melee.rs:177	can_advance_after_comb	175 // player's unit, not artillery, adjacent to `to`, no enemy in `to`, and 176 // the stacking law kept with any friendly units that advanced first. 177 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 178 let unit = self.unit_or_err(unit_id)?; 179 // §6.7: there is no advance after combat as a result of defensive fire.

Covered by tests: omdurman-rules::src::effects::tests::no_advance_after_defensive_fire
omdurman-rules::src::effects::tests::defensive_fire_opens_no_advance_window

§6.8 – Offensive Fire

descriptive
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Offensive Fire

§6.11 – Fire combat factor printed on units

implemented

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The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:18 GH:omdurman-rules/src/scalars.rs:18	FireFactor	<pre>16 // Every possible value from the annotated counter 17 set is a named variant. 18 #[derive(Serialize, Deserialize, Clone, Copy, 19 PartialEq, Eq, Hash, Debug, strum::Display)] 20 pub enum FireFactor { One = 1, Three = 3,</pre>

Covered by tests: [omdurman-rules::src::tests::fire_factor_sum_to_row](#)

§6.12 – Fire combat is always voluntary

implicit

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Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implemented

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If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

File	Symbol	Code Snippet
omdurman-rules/src/effects/fire.rs:668 GH:omdurman-rules/src/effects/fire.rs:668	build_fire_attack_from	<pre>666 // derived exactly as in [`build_fire_attack`]. 667 Returns `None` for an empty, 668 duplicated, or not-fully-legal list. 669 pub fn build_fire_attack_from(670 gs: &GameState, firer_hex: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::fire_rejects_duplicate_firers_and_a_foreign_firing_player](#)
[omdurman-rules::src::effects::tests::single_unit_fires_alone_from_a_stack](#)

§6.14 – Players may combine fire factors into one attack

implemented

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Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:32 GH:omdurman-rules/src/scalars.rs:32	sum_to_row	<pre> 30 impl FireFactor { 31 /// Sum multiple fire factors and return the corresponding Combat Results Table row (rulebook §6.11). 32 pub fn sum_to_row<'a>(factors: impl IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 33 let total: u16 = factors.into_iter().map(f f.value()).sum(); 34 crate::combat_results_table::FireFactorRow::from_total(total) </pre>
omdurman-rules/src/effects/state/fire.rs:51 GH:omdurman-rules/src/effects/state/fire.rs:51	GameS- tate::can_fire_at	<pre> 49 /// fire ignores it, §6.64), and that a gunboat, or a fort with nobody 50 /// inside, is fired at by artillery only (§6.61/ §6.62). 51 pub fn can_fire_at(52 &self, 53 firer: UnitId, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::brigade_integrity_survives_combined_fire](#)
[omdurman-rules::src::effects::tests::dead_units_are_pruned_from_every_tracker](#)
[omdurman-rules::src::effects::tests::unit_may_only_be_fired_at_once_per_phase](#)
[omdurman-rules::src::effects::tests::gunboat_and_maxim_may_be_fired_at_repeatedly](#)
[omdurman-rules::src::effects::tests::rejected_fire_attack_does_not_mark_firers_as_fired](#)
[omdurman-rules::src::effects::tests::single_unit_fires_alone_from_a_stack](#)
[omdurman-rules::src::effects::tests::infantry_may_fire_only_once_per_subphase](#)
[omdurman-rules::src::effects::tests::maxim_may_fire_twice_per_turn](#)
[omdurman-rules::src::effects::tests::fire_from_two_hexes_combines_into_one_attack](#)

§6.15 – May divide a stack to fire at different hexes

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Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/effects/fire.rs:668 GH:omdurman-rules/src/effects/fire.rs:668	build_fire_attack_from	<pre> 666 /// derived exactly as in [<code>build_fire_attack</code>]. Returns <code>None</code> for an empty, 667 /// duplicated, or not-fully-legal list. 668 pub fn build_fire_attack_from(669 gs: &GameState, 670 firer_hex: HexCoord, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::stack_may_split_its_fire_across_hexes](#)

§6.16 – Halving fire strength rounds down, minimum 1

implemented

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
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omdurman-rules/src/unit.rs:230 GH:omdurman-rules/src/unit.rs:230	RangeBand	<pre> 228 /// multiplied at a given distance (§6.22). 229 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 230 pub enum RangeBand { 231 Tripled, 232 Doubled,</pre>
---	------------------	---

Proven by: [omdurman-rules::src::verification::range_band_halved_is_max_of_one_and_floor_half](#)
[omdurman-rules::src::verification::range_band_apply_is_monotone_in_raw](#)
[omdurman-rules::src::verification::disrupt_half_is_rounded_up](#)

Covered by tests: [omdurman-rules::src::tests::halving_rounds_down_and_never_below_one](#)

§6.21 – First check LOS before firing

implemented

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When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:235 GH:omdurman-rules/src/los_table.rs:235	blocking_rules	<pre> 233 /// always blocks. Indexing is in-bounds by construction (both enums have 234 /// exactly three variants). 235 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 236 crate::tables_data::LOS_CELLS[firer.index()] [target.index()] 237 }</pre>
omdurman-rules/src/los_table.rs:584 GH:omdurman-rules/src/los_table.rs:584	has_los	<pre> 582 /// board-only tools). 583 #[allow(clippy::too_many_arguments)] 584 pub fn has_los(585 board: &crate::board::BoardInfo, 586 from: HexCoord,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::can_fire_at_rejects_blocked_los](#) [omdurman-rules::src::effects::tests::can_fire_at_allows_clear_los](#)

§6.22 – Consult Range Effects Table

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Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:34 GH:omdurman-rules/src/range_effects.rs:34	ae_range_effects	<pre> 32 /// Look up the range band for an Anglo-Egyptian weapon (§6.22, §6.24). 33 /// Distances > 10 are out of range for all weapons. 34 pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 35 band_at(faction_rows(true), weapon, distance) 36 }</pre>
omdurman-rules/src/range_effects.rs:40	dervish_range_effects	<pre> 38 /// Look up the range band for a Dervish weapon (§6.22).</pre>

GH:omdurman-rules/src/range_effects.rs:40		<pre> 39 /// Distances > 10 are out of range for all weapons. 40 pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 41 band_at(faction_rows(false), weapon, distance) 42 } </pre>
omdurman-rules/src/unit.rs:230 GH:omdurman-rules/src/unit.rs:230	RangeBand	<pre> 228 /// multiplied at a given distance (\$6.22). 229 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 230 pub enum RangeBand { 231 Tripled, 232 Doubled, </pre>
omdurman-rules/src/scalars.rs:126 GH:omdurman-rules/src/scalars.rs:126	HexDistance	<pre> 124 /// (rulebook \$6.22, \$7.5). 125 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 126 pub struct HexDistance(pub(crate) u16); 127 128 impl HexDistance { </pre>

Proven by:

- omdurman-types::src::lib::adjacency_iff_distance_one
- omdurman-rules::src::verification::fire_factor_row_from_total_matches_printed_bands
- omdurman-rules::src::verification::range_band_multiplier_arithmetic_is_exact
- omdurman-rules::src::range_effects::range_effects_are_out_of_range_outside_the_printed_ten_hex_distance_window
- omdurman-rules::src::range_effects::max_day_range_is_the_last_in_range_hex
- omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table

Covered by tests:

- omdurman-rules::src::effects::tests::numbered_result_elimimates_without_disrupting_survivors
- omdurman-rules::src::range_effects::ae_rifles_doubled_at_range_1
- omdurman-rules::src::range_effects::ae_rifles_halved_at_range_4
- omdurman-rules::src::range_effects::ae_howitzer_range
- omdurman-rules::src::range_effects::dervish_rifles_shorter_range
- omdurman-rules::src::range_effects::melee_only_range_1
- omdurman-rules::src::range_effects::ae_range_effects_artillery_full
- omdurman-rules::src::range_effects::ae_range_effects_maxims_match_rifles
- omdurman-rules::src::range_effects::ae_range_effects_distance_over_10
- omdurman-rules::src::range_effects::ae_range_effects_howitzer_halved_4_to_10
- omdurman-rules::src::range_effects::dervish_range_effects_rifles
- omdurman-rules::src::range_effects::dervish_range_effects_artillery
- omdurman-rules::src::range_effects::dervish_range_effects_maxims_and_howitzer
- omdurman-rules::src::range_effects::dervish_range_effects_melee
- omdurman-rules::src::range_effects::dervish_range_effects_distance_over_10
- omdurman-rules::src::effects::tests::fire_combat_elimimates_target
- omdurman-rules::src::range_effects::max_day_range_all_combos
- omdurman-rules::src::range_effects::range_effects_every_cell_ae_artillery
- omdurman-rules::src::range_effects::range_effects_every_cell_ae_howitzer
- omdurman-rules::src::range_effects::range_effects_every_cell_ae_maxims
- omdurman-rules::src::range_effects::range_effects_every_cell_ae_rifles
- omdurman-rules::src::range_effects::range_effects_every_cell_dervish_artillery
- omdurman-rules::src::range_effects::range_effects_every_cell_dervish_maxims_howitzer
- omdurman-rules::src::range_effects::range_effects_every_cell_dervish_rifles
- omdurman-rules::src::range_effects::range_effects_every_cell_dervish_spears
- omdurman-rules::src::effects::tests::can_fire_at_gates_phase_range_and_player
- omdurman-rules::src::effects::tests::mixed_attack_bands_per_firer
- omdurman-rules::src::range_effects::ae_range_effects_monotone_non_increasing
- omdurman-rules::src::range_effects::ae_howitzer_has_minimum_range
- omdurman-rules::src::range_effects::dervish_range_effects_monotone_non_increasing
- omdurman-rules::src::range_effects::range_effects_first_range_max_effect_last_range_oor
- omdurman-rules::src::effects::tests::a_disrupt_result_falls_on_undisrupted_units_first

§6.23 – Terrain defensive modifier

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre> 57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 } </pre>
omdurman-rules/src/combat.rs:26 GH:omdurman-rules/src/combat.rs:26	FireModifier::Terrain	<pre> 24 /// Negative modifier from the Terrain Effects Chart applied to the 25 /// defender's hex (§6.23). 26 Terrain(i16), 27 /// -2 thorn-hedge defensive modifier (§9.231). 28 ZaribaThornHedge, </pre>

Proven by: `omdurman-rules::src::verification::fire_modifier_keeps_roll_legal`
`omdurman-rules::src::terrain_chart::defence_column_never_helps_the_attacker`

Covered by tests:

- `omdurman-rules::src::terrain_chart::engine_matches_the_transcribed_terrain_effects_chart`
- `omdurman-rules::src::effects::tests::fire_across_a_crest_or_wall_is_penalised`
- `omdurman-rules::src::terrain_chart::clear_terrain_no_bonus`
- `omdurman-rules::src::terrain_chart::building_gives_minus_3`
- `omdurman-rules::src::terrain_chart::trees_cost_one_and_give_no_defense`
- `omdurman-rules::src::terrain_chart::rough_movement_and_defense`
- `omdurman-rules::src::terrain_chart::swamp_movement_and_defense`
- `omdurman-rules::src::terrain_chart::hilltop_movement_and_defense`
- `omdurman-rules::src::terrain_chart::huts_movement_and_defense`
- `omdurman-rules::src::terrain_chart::defense_modifier_convenience_matches_chart`
- `omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds`
- `omdurman-rules::src::terrain_chart::terrain_defense_modifier_non_positive`

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:20 GH:omdurman-rules/src/combat.rs:20	AngloEgyptianDirectFire	<pre> 18 pub enum FireModifier { 19 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 20 AngloEgyptianDirectFire, 21 /// +1 brigade integrity, applied only if all four battalions fire at 22 /// the same enemy-occupied hex (§5.54, §6.24). </pre>
omdurman-rules/src/unit.rs:626 GH:omdurman-rules/src/unit.rs:626	BrigadeIntegrity	<pre> 624 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 625 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 626 pub enum BrigadeIntegrity { 627 None, 628 Integrated(BrigadeId), </pre>
omdurman-rules/src/combat.rs:36 GH:omdurman-rules/src/combat.rs:36	FireModifier::die_modifier	<pre> 34 impl FireModifier { 35 /// Return the numeric die-roll modifier for this bonus/penalty (rulebook §6.24, §5.54, §6.23, §9.231, §9.232). 36 pub fn die_modifier(self) -> i16 { 37 match self { 38 FireModifier::AngloEgyptianDirectFire FireModifier::BrigadeIntegrity => 1, </pre>

Proven by: `omdurman-rules::src::verification::die_roll_apply_modifier_is_total`

Covered by tests: `omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived`
`omdurman-rules::src::effects::tests::fire_modifiers_are_engine_derived_and_mismatches_rejected`

§6.41 – Direct Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/turn.rs:46 GH:omdurman-rules/src/turn.rs:46	DirectFire	<pre>44 pub enum FireSubPhase { 45 /// Direct fire (§6.41). Both sides participate in this sub-phase. 46 DirectFire, 47 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 48 MaximSecondAndHowitzer,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::fire_rejects_duplicate_firers_and_a_foreign_firing_player`
`omdurman-rules::src::effects::tests::disrupted_unit_cannot_fire` `omdurman-rules::src::effects::tests::fire_combat_eliminate_target`
`omdurman-rules::src::effects::tests::walls_are_breached_in_the_direct_fire_subphase_only`

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/turn.rs:48 GH:omdurman-rules/src/turn.rs:48	MaximSecondAndHowitzer	<pre>46 DirectFire, 47 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 48 MaximSecondAndHowitzer, 49 } 50 </pre>
omdurman-types/src/lib.rs:1110 GH:omdurman-types/src/lib.rs:1110	fires_twice	<pre>1108 /// Maxim guns fire twice per turn -- once in the Direct Fire Subphase and 1109 /// again in the Maxim Second Fire Subphase (rulebook §6.42). 1110 pub fn fires_twice(self) -> bool { 1111 matches!(self, UnitKind::Maxim { .. }) 1112 }</pre>

Proven by: `omdurman-types::src::lib::unit_kind_fire_and_movement_capability_law_is_exact`

Covered by tests: `omdurman-rules::src::effects::tests::fall_of_khartoum_has_no_maxim_howitzer_subphase`
`omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10`
`omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7`
`omdurman-rules::src::effects::tests::advance_window_bridges_fire_subphase_and_closes_at_melee`
`omdurman-rules::src::effects::tests::fired_at_tracker_resets_at_maxim_subphase`
`omdurman-rules::src::effects::tests::maxim_may_fire_twice_per_turn`
`omdurman-rules::src::effects::tests::maxim_that_skipped_direct_may_fire_once_in_second_subphase`
`omdurman-rules::src::effects::tests::walls_are_breached_in_the_direct_fire_subphase_only`

§6.51 – Leader Units

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File	Symbol	Code Snippet
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omdurman-rules/src/unit.rs:115 GH:omdurman-rules/src/unit.rs:115	BritishLeader	<pre> 113 /// to claim the Mahdi's Tomb (§9.14). 114 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 115 pub enum BritishLeader { 116 Kitchener, 117 Gatacre, </pre>
omdurman-types/src/lib.rs:1057 GH:omdurman-types/src/lib.rs:1057	BritishLeader	<pre> 1055 }, 1056 /// Anglo-Egyptian leader (§6.51): movement only. 1057 BritishLeader { movement: i32 }, 1058 /// Wall-breach marker placed by artillery fire (\$6.63). Not a combat unit. 1059 Breech, </pre>
omdurman-types/src/lib.rs:1098 GH:omdurman-types/src/lib.rs:1098	has_combat_factors	<pre> 1096 /// British leaders print a movement factor only (\$6.51); other kinds carry 1097 /// fire and/or melee factors. Markers carry no stats. 1098 pub fn has_combat_factors(self) -> bool { 1099 !matches!(1100 self, </pre>

Proven by: [omdurman-types::src::lib::unit_kind_fire_and_movement_capability Law is exact](#)

Covered by tests:

- [omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero](#)
- [omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlys_camel_and_sudanese](#)
- [omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell](#)
- [omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup](#)
- [omdurman-rules::src::effects::tests::dervish_move_through_lone_ae_leader_hex_eliminate_the_leader](#)
- [omdurman-rules::src::effects::tests::dervish_move_onto_lone_ae_leader_hex_eliminate_the_leader](#)
- [omdurman-rules::src::effects::tests::ae_leader_with_combat_unit_is_not_overrun_by_dervish_move](#)
- [omdurman-rules::src::effects::tests::ae_leader_eliminated_with_last_combat_unit_in_fire_combat](#)
- [omdurman-rules::src::effects::tests::a_lone_gordon_makes_no_melee_roll](#)
- [omdurman-rules::src::effects::tests::melee_advance_overruns_a_lone_ae_leader](#)
- [omdurman-rules::src::effects::tests::entering_reinforcements_stop_in_zoc_and_overrun_lone_leaders](#)

§6.52 – Anglo-Egyptian Friendlys Brigade

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:339 GH:omdurman-rules/src/unit.rs:339	is_friendlys	<pre> 337 /// "Friendlys" units obey several special rules (\$5.21, §5.23, §6.52, 338 /// §9.14 victory conditions). 339 pub fn is_friendlys(&self) -> bool { 340 matches!(341 self, </pre>
omdurman-types/src/lib.rs:1140 GH:omdurman-types/src/lib.rs:1140	Friendlys	<pre> 1138 /// Native volunteer brigade -- the Shaggyeh (\$6.52). Do not receive 1139 /// brigade integrity (§5.54 enumerates only British/Egyptian/Sudanese). 1140 Friendlys, 1141 } 1142 </pre>

Proven by: [omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlys_and_fok_to_the_dervish_table](#)

Covered by tests:

- [omdurman-rules::src::unit_profiles::friendlys_counters_score_by_bank_not_as_leaders](#)
- [omdurman-rules::src::effects::tests::friendlys_validate_and_resolve_on_dervish_table](#)
- [omdurman-rules::src::effects::tests::friendlys_melee_with_the_dervish_modifier](#)

§6.53 – Royal Engineers demolition

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:308 GH:omdurman-rules/src/unit.rs:308	RoyalEngineers	<pre>306 /// The Royal Engineers (§6.53) -- a *specific* unit, not a class, so we 307 /// model it explicitly. 308 RoyalEngineers, 309 /// A British fort: FALL OF KHARTOUM's Forts Makran and Buri (§9.321). 310 AngloEgyptianFort,</pre>
omdurman-rules/src/unit.rs:551 GH:omdurman-rules/src/unit.rs:551	demolishing	<pre>549 /// Set when the Royal Engineers are committed to a demolition this turn 550 /// (§6.53) -- neither offensive fire nor melee allowed that turn. 551 pub demolishing: bool, 552 /// Set when a gunboat has lost its engines to a river mine (§10.12, roll 553 /// 5-7): it may no longer move under power and instead drifts two hexes per</pre>
omdurman-rules/src/effects/effect.rs:159 GH:omdurman-rules/src/effects/effect.rs:159	Demolition	<pre>157 158 /// Royal Engineers demolition (rulebook §6.53). 159 Demolition { 160 unit_id: UnitId, 161 target: DemolitionTarget,</pre>
omdurman-rules/src/effects/setup.rs:25 GH:omdurman-rules/src/effects/setup.rs:25	apply_demolition	<pre>23 /// resolution happens at end of turn ([`resolve_demolition`]), which checks 24 /// the engineer is still adjacent and undisrupted. 25 pub fn apply_demolition(26 state: &mut GameState, 27 unit_id: UnitId,</pre>
omdurman-rules/src/combat.rs:162 GH:omdurman-rules/src/combat.rs:162	DemolitionTarget	<pre>160 /// disrupted or driven off. 161 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 162 pub enum DemolitionTarget { 163 Fort(UnitId), 164 WallHexside(HexsideRef),</pre>
omdurman-rules/src/effects/state/engineering.rs:39 GH:omdurman-rules/src/effects/state/engineering.rs:39	demolition_targets	<pre>37 /// the rules would accept. Empty when the unit doesn't exist or has no 38 /// adjacent target. 39 pub fn demolition_targets(&self, unit_id: UnitId) - > Vec<DemolitionTarget> { 40 let Ok(unit) = self.unit_or_err(unit_id) else { 41 return Vec::new();</pre>

Covered by tests: [omdurman-rules::src::effects::tests::demolition_requires_engineers_an_adjacent_target_and_a_pending_commit](#)
[omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_disrupted](#)
[omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_moved_away](#)
[omdurman-rules::src::effects::tests::demolition_targets_finds_adjacent_fort_and_wall](#)
[omdurman-rules::src::effects::tests::a_demolished_fort_takes_an_occupant_with_it](#)

§6.54 – Forts

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/	ZocReason	<pre>582 /// Used by the engine when answering "is this hex in an enemy ZOC?".</pre>

unit.rs:584 GH:omdurman-rules/src/unit.rs:584		<pre> 583 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 584 pub enum ZocReason { 585 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 586 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-rules/src/unit.rs:591 GH:omdurman-rules/src/unit.rs:591	Fort	<pre> 589 GunboatVsGunboat, 590 /// Forts project ZOC out of, but not into, an empty fort (§5.44, §6.54). 591 Fort, 592 } 593 </pre>
omdurman-rules/src/unit.rs:539 GH:omdurman-rules/src/unit.rs:539	UnitState	<pre> 537 /// rather than one big enum. 538 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 539 pub struct UnitState { 540 /// Reference table: "Disrupted units: no ZOC; may not move; may not fire 541 /// offensively or defensively; may not melee; are turned face up at the </pre>
omdurman-rules/src/combat.rs:65 GH:omdurman-rules/src/combat.rs:65	FireAttack	<pre> 63 /// modifiers (rulebook §6). 64 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 65 pub struct FireAttack { 66 pub firing_player: Player, 67 pub phase: Phase, </pre>
omdurman-rules/src/combat.rs:87 GH:omdurman-rules/src/combat.rs:87	FireAttack::net_modifier	<pre> 85 impl FireAttack { 86 /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 87 pub fn net_modifier(&self) -> i16 { 88 // Saturating fold rather than `sum()`: the modifier list is unbounded 89 // and arrives over the network, so a plain sum can overflow `i16`. </pre>

Covered by tests: [omdurman-rules::src::effects::tests::printed_fok_forts_are_british_forts](#)
[omdurman-rules::src::board_data::fok_forts_are_not_walled_in](#)
[omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort](#)
[omdurman-rules::src::effects::tests::mandatory_advance_stops_at_an_enemy_fort](#)
[omdurman-rules::src::effects::tests::the_units_inside_a_fort_are_fired_at_with_its_minus_three](#)
[omdurman-rules::src::effects::tests::only_infantry_melee_destroys_a_fort](#)

§6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

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Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:187 GH:omdurman-rules/src/unit.rs:187	WeaponClass	<pre> 185 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 186)] 187 pub enum WeaponClass { 188 /// Dervish spears and swords -- no ranged fire at all. 189 Melee, </pre>
omdurman-rules/src/effects/state/fire.rs:51	GameState::can_fire_at	<pre> 49 /// fire ignores it, §6.64), and that a gunboat, or a fort with nobody 50 /// inside, is fired at by artillery only (§6.61/ </pre>

GH:omdurman-rules/src/effects/state/fire.rs:51	<pre>\$6.62). 51 pub fn can_fire_at(52 &self, 53 firer: UnitId,</pre>
--	--

Covered by tests: [omdurman-rules::src::effects::tests::a_gunboat_sunk_by_artillery_scores_victory_points](#)
[omdurman-rules::src::effects::tests::rifles_may_not_sink_a_gunboat](#)
[omdurman-rules::src::effects::tests::artillery_sinks_gunboat_only_on_three_plus](#)

\$6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

manual page unknown

Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/unit.rs:187 GH:omdurman-rules/src/unit.rs:187	WeaponClass	<pre>185 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 186)] 187 pub enum WeaponClass { 188 /// Dervish spears and swords -- no ranged fire at 189 all. 189 Melee,</pre>
omdurman-rules/src/effects/state/fire.rs:51 GH:omdurman-rules/src/effects/state/fire.rs:51	GameS- tate::can_fire_at	<pre>49 /// fire ignores it, §6.64), and that a gunboat, or 49 a fort with nobody 50 /// inside, is fired at by artillery only (§6.61/ 50 §6.62). 51 pub fn can_fire_at(52 &self, 53 firer: UnitId,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::a_destroyed_fort_takes_one_occupant_with_it_and_scores_it](#)
[omdurman-rules::src::effects::tests::artillery_destroys_fort_on_two_or_better_only](#)
[omdurman-rules::src::effects::tests::destroyed_fort_takes_one_occupant_with_it](#)
[omdurman-rules::src::effects::tests::the_units_inside_a_fort_are_fired_at_with_its_minus_three](#)
[omdurman-rules::src::effects::tests::a_demolished_fort_takes_an_occupant_with_it](#)

\$6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Subphase on the marchpost for the impact hex. The designated target hex is hit on a roll of 7–10. (see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:269 GH:omdurman-types/src/lib.rs:269	Breach	<pre> 267 /// Breach in a wall (artillery/\$6.63 or Royal Engineers/\$6.53). ZOC both 268 /// ways; LOS no longer blocked across the hexside. 269 Breach, 270 /// Khor -- gully/wadi. ZOCs do not extend across (\$5.44); advance after 271 /// combat may not cross (\$6.82). </pre>
omdurman-rules/src/effects/effect.rs:236 GH:omdurman-rules/src/effects/effect.rs:236	ArtilleryBreachWall	<pre> 234 /// pre-rolled d10 used for the CRT lookup; range/ LOS are re-derived by the 235 /// engine from the firers and `target`. 236 ArtilleryBreachWall { 237 firers: Vec<UnitId>, 238 target: HexsideRef, </pre>
omdurman-rules/src/effects/fire.rs:990 GH:omdurman-rules/src/effects/fire.rs:990	apply_artillery_breach_wall	<pre> 988 /// artillery's CRT roll -- the rulebook specifies the same "2+ required" 989 /// threshold for both trigger styles. 990 pub fn apply_artillery_breach_wall(991 state: &mut GameState, 992 firers: &[UnitId], </pre>
omdurman-rules/src/effects/state/fire.rs:258 GH:omdurman-rules/src/effects/state/fire.rs:258	can_fire_at_wall	<pre> 256 /// range band and resolving the CRT -- this method only validates one 257 /// firer at a time. 258 pub fn can_fire_at_wall(259 &self, 260 firer: UnitId, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::breach_casualty_stands_at_the_wall](#)
[omdurman-rules::src::unit_profiles::breach_marker_cell_returns_none](#)
[omdurman-rules::src::effects::tests::artillery_breaches_wall_only_on_crt_two_or_better](#)
[omdurman-rules::src::effects::tests::wall_breach_eliminate_one_adjacent_enemy_unit](#)
[omdurman-rules::src::effects::tests::breach_is_an_opening_for_los_and_effective_kinds](#)
[omdurman-rules::src::effects::tests::artillery_may_fire_at_the_wall_it_faces](#)
[omdurman-rules::src::effects::tests::walls_are_breached_in_the_direct_fire_subphase_only](#)

§6.64 – Howitzer fire

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:9 GH:omdurman-rules/src/howitzer_scatter.rs:9	ScatterHexDirection	<pre> 7 /// of the map. 8 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 9 pub enum ScatterHexDirection { 10 UpperLeft, 11 UpperRight, </pre>
omdurman-rules/src/howitzer_scatter.rs:56 GH:omdurman-rules/src/howitzer_scatter.rs:56	howitzer_scatter	<pre> 54 /// [`crate::combat_results_table`]); this function determines the *impact 55 /// hex* from the second roll, placed by [`scatter_impact_hex`]. 56 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterHexDirection { 57 58 crate::tables_data::SCATTERGRAM[(impact_roll.value() - 1) as usize] 59 } </pre>
omdurman-rules/src/unit.rs:126 GH:omdurman-rules/src/unit.rs:126	GunboatId	<pre> 124 /// fire; "old" gunboats do not (rulebook §2.32). 125 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 126 pub enum GunboatId { 127 /// One of the five new-type named gunboats with </pre>

		<pre> howitzer capability. 128 Named(NamedGunboat), </pre>
omdurman-rules/src/effects/effect.rs:89 GH:omdurman-rules/src/effects/effect.rs:89	HowitzerFire	<pre> 87 /// - CRT result applied to units at impact hex (not the original target). 88 /// - Firers marked as fired. 89 HowitzerFire { 90 attack: FireAttack, 91 combat_results_table_roll: DieRoll, </pre>
omdurman-rules/src/effects/fire.rs:18 GH:omdurman-rules/src/effects/fire.rs:18	apply_howitzer_fire	<pre> 16 17 /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 18 pub fn apply_howitzer_fire(19 state: &mut GameState, 20 attack: &FireAttack, </pre>
omdurman-rules/src/effects/state/fire.rs:51 GH:omdurman-rules/src/effects/state/fire.rs:51	can_fire_at	<pre> 49 /// fire ignores it, §6.64), and that a gunboat, or a fort with nobody 50 /// inside, is fired at by artillery only (§6.61/ §6.62). 51 pub fn can_fire_at(52 &self, 53 firer: UnitId, </pre>

Proven by: [omdurman-rules::src::effects::scatter_lands_on_or_next_to_the_target](#)
[omdurman-rules::src::howitzer_scatter::scatter_is_center_exactly_for_rolls_7_to_10](#)

Covered by tests: [omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10](#)
[omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7](#)
[omdurman-rules::src::howitzer_scatter::howitzer_each_miss_gets_its_printed_hex](#)
[omdurman-rules::src::unit_profiles::named_and_old_gunboats_resolve](#)
[omdurman-rules::src::effects::tests::named_gunboat_has_howitzer](#)
[omdurman-rules::src::effects::tests::a_scattered_howitzer_shell_hits_a_friendly_occupied_hex](#)
[omdurman-rules::src::effects::tests::a_rejected_howitzer_shot_leaves_no_impact_record](#)
[omdurman-rules::src::effects::tests::named_gunboat_may_fire_howitzer_in_second_subphase](#)
[omdurman-rules::src::effects::tests::named_gunboat_direct_fire_uses_artillery_weapon](#)
[omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night](#)
[omdurman-rules::src::effects::tests::dervish_gunboat_lacks_howitzer](#)
[omdurman-rules::src::howitzer_scatter::scatter_follows_the_printed_diagram](#)

§6.81 – Moving player may fire with all capable units

implicit

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During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

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If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:146	AdvanceAfterCombat	<pre> 144 /// **Postconditions:** Unit position moved to `to`; `vacated_by_combat` </pre>

GH:omdurman-rules/src/effects/effect.rs:146		<pre> 145 /// entry consumed. 146 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 147 148 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects/dispatch.rs:40 GH:omdurman-rules/src/effects/dispatch.rs:40	apply_advance_after_combat	<pre> 38 } 39 GameEffect::AdvanceAfterCombat { unit_id, to } => { 40 apply_advance_after_combat(state, *unit_id, *to) 41 } 42 GameEffect::ConstructZariba { unit_ids, hexside } => { </pre>
omdurman-types/src/lib.rs:319 GH:omdurman-types/src/lib.rs:319	blocks_advance_after_combat	<pre> 317 318 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 319 pub fn blocks_advance_after_combat(self) -> bool { 320 matches!(321 self, </pre>
omdurman-rules/src/effects/state/melee.rs:177 GH:omdurman-rules/src/effects/state/melee.rs:177	can_advance_after_combat	<pre> 175 /// player's unit, not artillery, adjacent to `to`, no enemy in `to`, and 176 /// the stacking law kept with any friendly units that advanced first. 177 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 178 let unit = self.unit_or_err(unit_id)?; 179 // §6.7: there is no advance after combat as a result of defensive fire. </pre>

Proven by: `omdurman-rules::src::effects::advance_phase_is_atomic` `omdurman-types::src::lib::hexside_blocking_classifiers_are_exact`

Covered by tests:

- `omdurman-app::src::melee::one_click_advances_the_stack_up_to_the_stacking_limit`
- `omdurman-rules::src::effects::tests::several_units_advance_into_one_vacated_hex`
- `omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_wall_hexside`
- `omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_khor_hexside`
- `omdurman-rules::src::effects::tests::advance_requires_combat_vacated_hex`
- `omdurman-rules::src::effects::tests::advance_requires_participation`
- `omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows`
- `omdurman-rules::src::effects::tests::a_gunboat_is_not_offered_a_land_advance`

7/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

descriptive

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Melee Phase

§7.1 – Melee strength printed on counter

implemented

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The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:42 GH:omdurman-rules/src/scalars.rs:42	MeleeFactor	<pre>40 // Every possible value from the annotated counter set is a named variant. 41 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 42 pub enum MeleeFactor { 43 One = 1, 44 Three = 3,</pre>
omdurman-rules/src/scalars.rs:53 GH:omdurman-rules/src/scalars.rs:53	MeleeFactor::sum	<pre>51 impl MeleeFactor { 52 // Sum multiple melee factors (rulebook §7.1). 53 pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 { 54 factors.into_iter().map(f f.value()).sum() 55 }</pre>
omdurman-types/src/lib.rs:1081 GH:omdurman-types/src/lib.rs:1081	may_be_melee_attacked	<pre>1079 1080 // Gunboats neither attack nor are attacked in melee (§7.1). 1081 pub fn may_be_melee_attacked(self) -> bool { 1082 !matches!(self, UnitKind::Gunboat { .. }) 1083 }</pre>

Proven by: [omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::melee_rejects_duplicate_attackers_and_partial_defenders](#)
[omdurman-rules::src::effects::tests::single_shot_melee_is_validated_like_a_declaration](#)
[omdurman-rules::src::effects::tests::a_path_may_not_pass_through_an_enemy_hex](#)
[omdurman-rules::src::tests::melee_factor_values_and_sum](#) [omdurman-rules::src::tests::unit_kind_melee_capability](#)

§7.2 – Melee adjacent only, not across wall hexsides

implemented

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Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:308	blocks_melee	<pre>306 // Effects Chart, Khor: "May not melee across"). Gates and breaches are 307 // passable to melee.</pre>

GH:omdurman-types/src/lib.rs:308		<pre> 308 pub fn blocks_melee(self) -> bool { 309 matches!(310 self, </pre>
omdurman-rules/src/effects/state/melee.rs:29 GH:omdurman-rules/src/effects/state/melee.rs:29	can_melee	<pre> 27 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 28 /// thorn-hedge hexside blocks the attack (§7.2). 29 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 30 let unit = self.unit_or_err(attacker)?; 31 </pre>

Proven by: `omdurman-types::src::lib::hexside_blocking_classifiers_are_exact`

Covered by tests: `omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind`
`omdurman-rules::src::effects::tests::can_melee_rejects_wall_hexside`
`omdurman-rules::src::effects::tests::can_melee_rejects_thorn_hedge_hexside`
`omdurman-rules::src::effects::tests::can_melee_allows_gate_hexside`

§7.3 – Simultaneous melee combat

implemented

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Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:119 GH:omdurman-rules/src/effects/effect.rs:119	ResolveMelee	<pre> 117 /// Results Table (§7.3, §7.7), the Dervish advance forced and the 118 /// Anglo-Egyptian one offered (§7.6); `pending_melee` cleared. 119 ResolveMelee, 120 121 /// A cavalry/camel unit retreats two hexes from an impending infantry </pre>
omdurman-rules/src/effects/melee.rs:358 GH:omdurman-rules/src/effects/melee.rs:358	apply_resolve_melee	<pre> 356 /// target hex (so a retreated defender is spared), then clear the window 357 /// (§7.5). 358 pub fn apply_resolve_melee(state: &mut GameState) -> Result<(), RuleError> { 359 let Some(pending) = state.pending_melee.as_ref() else { 360 return Err(RuleError::NoMeleePending); </pre>

Covered by tests: `omdurman-rules::src::effects::tests::melee_resolves_simultaneously`

§7.4 – Who may melee attack / defend

implemented

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Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:1070 GH:omdurman-	may_melee_attack	<pre> 1068 /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may 1069 /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1). </pre>

types/src/ lib.rs:1070		1070 pub fn may_melee_attack(self) -> bool { 1071 matches!(1072 self,
omdurman- types/src/ lib.rs:1007 GH:omdurman- types/src/ lib.rs:1007	UnitKind	1005 /// `Some(UnitKind::Marker)` or `None`. 1006 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 1007 pub enum UnitKind { 1008 /// Foot infantry (§2.3): fire / melee / movement. 1009 Infantry {
omdurman- types/src/ lib.rs:840 GH:omdurman- types/src/lib.rs:840	DervishTribe	838 strum::EnumIter, 839 }] 840 pub enum DervishTribe { 841 Baggara, 842 Jaalin,
omdurman- rules/src/effects/ state/melee.rs:29 GH:omdurman- rules/src/effects/ state/melee.rs:29	can_melee	27 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 28 /// thorn-hedge hexside blocks the attack (§7.2). 29 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 30 let unit = self.unit_or_err(attacker)?; 31

Proven by: [omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind](#)

§7.5 – Cavalry/camel retreat before melee

implemented

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Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman- rules/src/effects/ effect.rs:132 GH:omdurman- rules/src/effects/ effect.rs:132	RetreatBeforeMelee	130 /// 131 /// **Postconditions:** Unit position moved to `to`. 132 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 133 134 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82
omdurman- rules/src/effects/ dispatch.rs:37 GH:omdurman- rules/src/effects/ dispatch.rs:37	apply_retreat_before_melee	35 GameEffect::ResolveMelee => apply_resolve_melee(state), 36 GameEffect::RetreatBeforeMelee { unit_id, to } => { 37 apply_retreat_before_melee(state, *unit_id, *to) 38 } 39 GameEffect::AdvanceAfterCombat { unit_id, to } => {
omdurman- types/src/ lib.rs:1087 GH:omdurman- types/src/ lib.rs:1087	may_retreat_before_melee	1085 /// Cavalry and camel units may retreat two hexes from an infantry melee 1086 /// attack (§7.5). 1087 pub fn may_retreat_before_melee(self) -> bool { 1088 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. }) 1089 }

omdurman-rules/src/scalars.rs:126 GH:omdurman-rules/src/scalars.rs:126	HexDistance	<pre> 124 /// (rulebook §6.22, §7.5). 125 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 126 pub struct HexDistance(pub(crate) u16); 127 128 impl HexDistance { </pre>
omdurman-rules/src/effects/state/melee.rs:98 GH:omdurman-rules/src/effects/state/melee.rs:98	can_retreat_before_melee	<pre> 96 /// away, free of the enemy and within the stacking law, reached through 97 /// an open intervening hex. 98 pub fn can_retreat_before_melee(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 99 let unit = self.unit_or_err(unit_id)?; 100 if !matches!(self.phase, Phase::Melee) { </pre>

Proven by: omdurman-rules::src::effects::resolve_melee_is_atomic omdurman-rules::src::effects::advance_phase_is_atomic
omdurman-types::src::lib::unit_kind_melee_capability_law_is_exact

Covered by tests: omdurman-rules::src::effects::tests::melee_lapses_when_every_defender_withdrew
omdurman-rules::src::effects::tests::retreat_before_melee_only_cavalry_two_hexes
omdurman-rules::src::effects::tests::retreat_opens_window_only_when_hex_emptyies
omdurman-rules::src::effects::tests::rejected_resolve_melee_keeps_the_declaration
omdurman-rules::src::effects::tests::a_retreat_needs_an_open_path_and_may_join_friends
omdurman-rules::src::effects::tests::a_unit_melee_attacks_once_a_turn

§7.6 – Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units MUST advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:146 GH:omdurman-rules/src/effects/effect.rs:146	AdvanceAfterCombat	<pre> 144 /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 145 /// entry consumed. 146 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 147 148 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects/dispatch.rs:40 GH:omdurman-rules/src/effects/dispatch.rs:40	apply_advance_after_combat	<pre> 38 } 39 GameEffect::AdvanceAfterCombat { unit_id, to } => { 40 apply_advance_after_combat(state, *unit_id, *to) 41 } 42 GameEffect::ConstructZariba { unit_ids, hexside } => { </pre>
omdurman-rules/src/effects/state/melee.rs:177 GH:omdurman-rules/src/effects/state/melee.rs:177	can_advance_after_combat	<pre> 175 /// player's unit, not artillery, adjacent to `to`, no enemy in `to`, and 176 /// the stacking law kept with any friendly units that advanced first. 177 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 178 let unit = self.unit_or_err(unit_id)?; 179 // §6.7: there is no advance after combat as a result of defensive fire. </pre>
omdurman-rules/src/effects/	apply_resolve_melee	<pre> 356 /// target hex (so a retreated defender is spared), then clear the window </pre>

melee.rs:358 GH:omdurman-rules/src/effects/melee.rs:358	<pre> 357 /// (\$7.5). 358 pub fn apply_resolve_melee(state: &mut GameState) -> Result<(), RuleError> { 359 let Some(pending) = state.pending_melee.as_ref() else { 360 return Err(RuleError::NoMeleePending); </pre>
--	--

Proven by: [omdurman-rules::src::effects::advance_phase_is_atomic](#)

Covered by tests:

- [omdurman-rules::src::effects::tests::melee_lapses_when_every_defender_withdrew](#)
- [omdurman-app::src::melee::one_click_advances_the_stack_up_to_the_stacking_limit](#)
- [omdurman-rules::src::effects::tests::several_units_advance_into_one_vacated_hex](#)
- [omdurman-rules::src::effects::tests::advancing_consumes_the_units_eligibility](#)
- [omdurman-rules::src::effects::tests::dervish_advance_after_melee_is_mandatory](#)
- [omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows](#)
- [omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking](#)
- [omdurman-rules::src::effects::tests::a_lone_gordon_makes_no_melee_roll](#)
- [omdurman-rules::src::effects::tests::melee_advance_overruns_a_lone_ae_leader](#)
- [omdurman-rules::src::effects::tests::mandatory_advance_stops_at_an_enemy_fort](#)

§7.7 – Melee modifiers

implemented

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To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:115 GH:omdurman-rules/src/combat.rs:115	MeleeModifier	<pre> 113 114 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 115 pub enum MeleeModifier { 116 /// +2 to all Dervish melee rolls (§7.7). 117 DervishStandard, </pre>
omdurman-rules/src/effects/melee.rs:462 GH:omdurman-rules/src/effects/melee.rs:462	mandatory_melee_modifiers	<pre> 460 /// `(attacker_modifiers, defender_modifiers)`; the engine applies exactly 461 /// these at resolution and rejects a declared attack whose lists differ. 462 pub fn mandatory_melee_modifiers(463 state: &GameState, 464 attack: &MeleeAttack, </pre>
omdurman-rules/src/combat.rs:143 GH:omdurman-rules/src/combat.rs:143	MeleeAttack	<pre> 141 /// A melee attack: simultaneous, both sides roll on the Combat Results Table (§7.3, §7.7). 142 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 143 pub struct MeleeAttack { 144 pub attacker_player: Player, 145 pub attacker_hex: HexCoord, </pre>
omdurman-rules/src/combat.rs:119 GH:omdurman-rules/src/combat.rs:119	MeleeModifier::Anglo-EgyptianStandard	<pre> 117 DervishStandard, 118 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 119 AngloEgyptianStandard, 120 /// Inverted to -2 when Dervish units melee-attack across a trench into 121 /// an entrenched defender (§9.232). </pre>
omdurman-rules/src/combat.rs:122	MeleeModifier::DervishVsDefender	<pre> 120 /// Inverted to -2 when Dervish units melee-attack across a trench into </pre>

GH:omdurman-rules/src/combat.rs:122		121 /// an entrenched defender (§9.232). 122 DervishVsTrenchedDefender, 123 /// +2: the Anglo-Egyptian "Friendlies" melee with the Dervish melee 124 /// modifier (§6.52) -- in place of the +1 standard when every
omdurman-rules/src/combat.rs:117 GH:omdurman-rules/src/combat.rs:117	MeleeModifier::DervishStandard	115 pub enum MeleeModifier { 116 /// +2 to all Dervish melee rolls (§7.7). 117 DervishStandard, 118 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 119 AngloEgyptianStandard,

Proven by: omdurman-rules::src::verification::melee_modifier_keeps_roll_legal

Covered by tests: omdurman-rules::src::effects::tests::melee_resolves_simultaneously
omdurman-rules::src::effects::tests::melee_modifiers_are_engine_derived_and_mismatches_rejected
omdurman-rules::src::effects::tests::friendlies_melee_with_the_dervish_modifier
omdurman-rules::src::effects::tests::disrupted_defenders_add_no_melee_factor
omdurman-rules::src::effects::tests::melee_losses_beyond_the_attackers_fall_on_their_stack

2/3 implemented

§8 – Night Game Turns

§8 – Night Game Turns

descriptive

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Night Game Turns

§8.1 – Night effects

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The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/scalars.rs:88 GH:omdurman-rules/src/scalars.rs:88	MovementAllowance::halve	<pre>86 impl MovementAllowance { 87 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 88 pub fn halve(self) -> Self { 89 let v = self.value() / 2; 90 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:907 GH:omdurman-types/src/lib.rs:907	DayNight	<pre>905 /// (rulebook §8.1). 906 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 907 pub enum DayNight { 908 Day, 909 Night,</pre>
omdurman-rules/src/range_effects.rs:71 GH:omdurman-rules/src/range_effects.rs:71	night_max_range	<pre>69 70 /// The halved maximum range at night (§8.1): round down, minimum 1. 71 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 72 let day = max_day_range(weapon, ae); 73 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/scalars.rs:208 GH:omdurman-rules/src/scalars.rs:208	effective_movement_at_night	<pre>206 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 207 /// Anglo-Egyptian movement allowances are halved (round down). 208 pub fn effective_movement_at_night(209 allowance: MovementAllowance, 210 player: Player,</pre>
omdurman-rules/src/effects/	NoHowitzerAtNight	<pre>112 // §6.64: no howitzer fire at night. 113 if kind == FireKind::Howitzer && self.day_night</pre>

state/fire.rs:114 GH:omdurman-rules/src/effects/state/fire.rs:114		<pre> == DayNight::Night { 114 return Err(RuleError::NoHowitzerAtNight); 115 } 116 </pre>
--	--	---

Proven by: `omdurman-rules::src::verification::movement_allowance_halve_never_panics`
`omdurman-rules::src::verification::night_halving_is_ac_only_and_day_neutral`
`omdurman-rules::src::range_effects::night_cap_is_halved_day_max_floored_at_one`
`omdurman-rules::src::range_effects::night_range_effects_gates_the_day_band_by_the_cap`
`omdurman-rules::src::effects::fire::night_capped_distance_is_some_exactly_within_the_night_max`

Covered by tests: `omdurman-rules::src::range_effects::night_max_ranges` `omdurman-rules::src::range_effects::night_max_ranges_remaining`
`omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example`
`omdurman-rules::src::range_effects::max_day_range_all_combos`
`omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved`
`omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night`
`omdurman-rules::src::effects::tests::british_gunboats_are_halved_at_night`

§8.2 – Dervish Desertion Roll

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:174 GH:omdurman-rules/src/effects/effect.rs:174	DervishDesertion	<pre> 172 /// the effect. The Khalifa, gunboats, artillery, and forts may not be 173 /// chosen. 174 DervishDesertion { 175 roll: DieRoll, 176 deserters: Vec<UnitId>, </pre>
omdurman-rules/src/unit.rs:389 GH:omdurman-rules/src/unit.rs:389	is_desertion_exempt	<pre> 387 /// Non-Dervish identities are trivially not eligible to desert and so are 388 /// reported as exempt too. 389 pub fn is_desertion_exempt(&self) -> bool { 390 match self { 391 UnitIdentity::DervishLeader(DervishLeader::KhalifaAbdullah) </pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre> 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. 63 DervishDesertion, 64 } 65 </pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre> 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. 63 DervishDesertion, 64 } 65 </pre>
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre> 58 /// Special events that occur on specific turns (rulebook §8.2, §9.112, §9.113). 59 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 60 pub enum TurnEvent { 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. </pre>

Proven by: `omdurman-rules::src::effects::desertion_count_is_one_and_a_half_times_the_roll`
`omdurman-rules::src::turn_track::desertion_event_is_unique_to_the_first_night_turn`

Covered by tests: omdurman-rules::src::effects::tests::desertion_rejects_a_unit_listed_twice
omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half
omdurman-rules::src::turn_track::desertion_on_first_night
omdurman-rules::src::effects::tests::desertion_roll_required_before_first_night_movement_ends
omdurman-rules::src::effects::tests::desertion_removes_chosen_count_and_respects_exemptions
omdurman-rules::src::effects::tests::desertion_is_rolled_in_the_dervish_movement_phase

23/33 implemented

§9 – The Scenarios

§9 – The Scenarios

descriptive

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The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

§9.2 – The Historical Scenario

descriptive

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The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa’s generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

§9.3 – Bonus Game: Fall of Khartoum

descriptive

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Bonus Game: FALL OF KHARTOUM

§9.11 – Set Up (Campaign)

descriptive

manual page unknown

Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-	GameTime	<pre>9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11 pub enum GameTime {</pre>

rules/src/ turn_track.rs:11		12 SixAM, 13 EightAM,
omdurman- rules/src/ turn_track.rs:47 GH:omdurman- rules/src/ turn_track.rs:47	TurnEntry	45 /// A single entry on the Turn Record Track (rulebook §9.12, §9.22). 46 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,
omdurman- rules/src/ turn_track.rs:91 GH:omdurman- rules/src/ turn_track.rs:91	CAMPAIGN_TURN_TRACK	89 /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) -- 90 /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell. 91 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [92 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT. 93 entry(1, SixAM, Day, TurnEvent::None),
omdurman- rules/src/turn_track.rs:239 GH:omdurman- rules/src/ turn_track.rs:239	TurnLabel	237 /// printed cell; `Blank` is for unused positions in the 9×3 grid. 238 #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)] 239 pub enum TurnLabel { 240 Blank, 241 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).

Proven by: omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario
omdurman-rules::src::turn_track::turn_labels_agree_with_the_rule_bearing_track

Covered by tests: omdurman-rules::src::turn_track::campaign_track_22_turns omdurman-rules::src::turn_track::desertion_on_first_night
omdurman-rules::src::turn_track::campaign_track_label_and_day_night_agree
omdurman-rules::src::turn_track::game_time_display_all_variants omdurman-rules::src::turn_track::turn_label_display
omdurman-rules::src::turn_track::turn_label_out_of_range_is_none
omdurman-rules::src::effects::tests::game_over_after_campaign_turns

§9.13 – Special Rules (Campaign)

descriptive
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Special Rules

None.

§9.14 – Victory Conditions (Campaign)

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Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

****Dervish Player receives:****

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/victory.rs:16 GH:omdurman-rules/src/victory.rs:16	VpSource	<pre> 14 /// the manual and the engine. 15 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 16 pub enum VpSource { 17 // ----- Anglo-Egyptian player receives: 18 /// 25 pts -- the Mahdi's Tomb taken from the Dervish: held at the </pre>
omdurman-rules/src/victory.rs:48 GH:omdurman-rules/src/victory.rs:48	VpSource::points	<pre> 46 impl VpSource { 47 /// VP awarded to `who_scores()` (rulebook §9.14). 48 pub fn points(self) -> VictoryPoints { 49 match self { 50 VpSource::MahdisTombTaken VpSource::MahdisTombHeld => VictoryPoints::new(25), </pre>
omdurman-rules/src/victory.rs:63 GH:omdurman-rules/src/victory.rs:63	VpSource::who_scores	<pre> 61 62 /// Which player receives these victory points (rulebook §9.14). 63 pub fn who_scores(self) -> Player { 64 match self { 65 VpSource::MahdisTombTaken </pre>
omdurman-rules/src/victory.rs:100 GH:omdurman-rules/src/victory.rs:100	VictoryLedger	<pre> 98 /// Cumulative victory ledger for one scenario (rulebook §9.14). 99 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 100 pub struct VictoryLedger { 101 pub events: Vec<VpEvent>, 102 } </pre>
omdurman-rules/src/victory.rs:106 GH:omdurman-rules/src/victory.rs:106	VpEvent	<pre> 104 /// A single victory-point scoring event (rulebook §9.14). 105 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 106 pub struct VpEvent { 107 pub turn: GameTurnIndex, 108 pub source: VpSource, </pre>
omdurman-rules/src/victory.rs:113 GH:omdurman-rules/src/victory.rs:113	VictoryLedger::total_for	<pre> 111 impl VictoryLedger { 112 /// Total victory points earned by a given player (rulebook §9.14). 113 pub fn total_for(&self, player: Player) -> VictoryPoints { 114 VictoryPoints(115 self.events </pre>
omdurman-rules/src/victory.rs:125 GH:omdurman-rules/src/victory.rs:125	VictoryLedger::superiority	<pre> 123 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 124 /// (rulebook §9.14). 125 pub fn superiority(&self) -> VictoryPoints { 126 VictoryPoints(127 self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(), </pre>
omdurman-rules/src/victory.rs:162 GH:omdurman-rules/src/victory.rs:162	CampaignVictoryLevel	<pre> 160 /// Campaign-game victory levels (§9.14). 161 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 162 pub enum CampaignVictoryLevel { 163 Draw, 164 Marginal(Player), </pre>
omdurman-rules/src/victory.rs:171 GH:omdurman-rules/src/victory.rs:171	CampaignVictoryLevel::from	<pre> 169 impl CampaignVictoryLevel { 170 /// Assign a level from the net superiority (§9.14). 171 pub fn from_superiority(s: VictoryPoints) -> Self { 172 let net = s.0; 173 // Positive -> Anglo-Egyptian thresholds: 15/30/50 </pre>

omdurman-rules/src/effects/victory.rs:75 GH:omdurman-rules/src/effects/victory.rs:75	score_elimination	<pre>73 /// elimination under `cause`. The owner is derived 74 /// so unlike the historical signature there is no 75 /// caller-supplied player. 76 pub fn score_elimination(state: &mut GameState, 77 unit_id: UnitId, cause: ElimCause) { 78 if let Some(unit) = state.find_unit(unit_id) { 79 let identity = unit.profile.identity;</pre>
omdurman-rules/src/scalars.rs:178 GH:omdurman-rules/src/scalars.rs:178	VictoryPoints	<pre>176 /// (rulebook §9.14). 177 #[derive(Serialize, Deserialize, Clone, Copy, 178 PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 179 pub struct VictoryPoints(pub(crate) i32); 180 impl VictoryPoints {</pre>

Proven by: [omdurman-rules::src::verification::campaign_victory_levels_match_manual_superiority_table](#)
[omdurman-rules::src::verification::vp_source_points_and_scorer_match_the_printed_schedule](#)
[omdurman-rules::src::effects::victory::vp_source_for_routes_every_elimination_to_the_printed_source](#)
[omdurman-rules::src::effects::victory::score_elimination_records_exactly_what_it_scores](#)

Covered by tests: [omdurman-rules::src::effects::tests::a_destroyed_fort_takes_one_occupant_with_it_and_scores_it](#)
[omdurman-rules::src::effects::tests::a_mined_gunboat_scores_and_takes_its_friendlies_down](#)
[omdurman-rules::src::effects::tests::a_gunboat_sunk_by_artillery_scores_victory_points](#)
[omdurman-rules::src::effects::tests::friendlies_bank_scores_by_side](#)
[omdurman-rules::src::effects::tests::mahdis_tomb_stays_dervish_without_a_leader](#)
[omdurman-rules::src::effects::tests::mahdis_tomb_scores_for_anglo_egyptian_when_taken](#)
[omdurman-rules::src::effects::tests::campaign_end_scores_an_untaken_tomb_for_the_dervish](#)
[omdurman-rules::src::tests::vp_source_attributes](#)

§9.21 – Set Up (Historical)

[descriptive](#)
manual page unknown

Set Up

§9.22 – Scenario Length (Historical)

[implemented](#)
manual page unknown

Scenario Length
6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:128 GH:omdurman-rules/src/turn_track.rs:128	HISTORICAL_TURN	<pre>126 127 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 128 6:00 am -> 12:00 pm). 129 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [130 TurnEntry { 131 turn: 1,</pre>

Proven by: [omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario](#)

Covered by tests: [omdurman-rules::src::turn_track::historical_turn_all_four_turns](#)

§9.23 – Special Rule: The Zariba

[descriptive](#)
manual page unknown

Special Rule: “The Zariba”

9.231) Thorn hedge hexsides: –2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

9.232) Trench hexsides: –4 to die roll on all Dervish fire attacks vs. entrenched units only; –2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented

manual page unknown

Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/victory.rs:201 GH:omdurman-rules/src/victory.rs:201	HistoricalVictoryLevel	<pre>199 /// draw"). 200 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 201 pub enum HistoricalVictoryLevel { 202 Draw = 1, 203 Marginal = 2,</pre>

Proven by: [omdurman-rules::src::verification::historical_victory_ladders_match_manual_bands](#)

Covered by tests: [omdurman-rules::src::tests::historical_victory_level_for_dervish](#)
[omdurman-rules::src::tests::historical_victory_level_for_anglo_egyptian](#)
[omdurman-rules::src::tests::historical_net_result_follows_the_worked_example](#)
[omdurman-rules::src::newspaper::historical_headline_follows_the_net_result](#)
[omdurman-rules::src::turn_summary::only_the_campaign_reports_victory_points](#)

§9.31 – Bonus game map

implemented

manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.

File	Symbol	Code Snippet
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omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	fall_of_khartoum_map	<pre>35 36 37 38 39 </pre>
---	-----------------------------	---

Covered by tests: [omdurman-app::src::tests::scenario_maps_to_board](#) [omdurman-app::src::tests::start_game_scenario_selects_board](#)

§9.32 – Set Up (Bonus)

descriptive
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

implemented
manual page unknown

Scenario Length

Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:169 GH:omdurman-rules/src/turn_track.rs:169	FALL_OF_KHARTOUM	<pre>167 168 169 170 171 </pre>

Proven by: [omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turn_one](#)

Covered by tests: [omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_turns_2_to_8_are_day](#)

§9.34 – Special Rules (Bonus)

descriptive
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

implemented
manual page unknown

Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/victory.rs:375 GH:omdurman-rules/src/victory.rs:375	FoKVictoryLevel::resolve	<pre> 373 /// rulebook: GORDON dies turn 5 (tactical) with 24 Dervish losses (-2 374 /// levels) nets a British marginal. 375 pub fn resolve(gordon_died_turn: Option<u8>, scenario_end_turn: u8, dervish_lost: i16) -> Self { 376 let base = Self::base(gordon_died_turn, scenario_end_turn); 377 let base_idx = Self::LADDER </pre>

Proven by: [omdurman-rules::src::verification::fok_victory_ladder_penalties_shift_monotonically](#)

Covered by tests: [omdurman-rules::src::effects::tests::gordon_dying_in_melee_ends_fall_of_khartoum](#)
[omdurman-rules::src::tests::fok_victory_level_worked_example](#) [omdurman-rules::src::tests::fok_victory_level_gordon_died_early](#)
[omdurman-rules::src::tests::fok_victory_level_late_gordon_death](#) [omdurman-rules::src::tests::fok_victory_level_gordon_survived](#)
[omdurman-rules::src::newspaper::fok_headlines_never_contradict_gordons_fate](#)

§9.111 – Dervish set up (Campaign)

implemented

manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taisha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112)** Dervish reinforcements: all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/setup.rs:143 GH:omdurman-rules/src/effects/state/setup.rs:143	setup_complete	<pre> 141 /// Returns [<code>RuleError::SetupIncomplete</code>] naming the first unmet requirement, 142 /// so the UI can surface *why* "Begin battle" is disabled. 143 pub fn setup_complete(&self) -> Result<(), RuleError> { 144 // Every scenario pins both orders of battle (§9.111/§9.113, 145 // §9.211-9.212, §9.321-9.322) -- in the Campaign the A-E side starts </pre>

Covered by tests: [omdurman-rules::src::effects::tests::setup_is_sequential_first_side_then_second](#)
[omdurman-app::src::scenario_setup::campaign_has_no_fixed_placements](#)
[omdurman-rules::src::unit_profiles::hadendowa_first_cell_is_isa_zachneih](#)
[omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

omdurman-rules::src::unit_profiles::dervish_roster_matches_the_order_of_appearance
omdurman-rules::tests::campaign_order_of_battle::the_dervish_initial_force_sets_up_where_printed

§9.112 – Dervish reinforcements (Campaign)

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:166 GH:omdurman-rules/src/effects/effect.rs:166	PlaceReinforcements	<pre> 164 // -- Reinforcement / placement ----- 165 /// Place reinforcements onto the map (rulebook \$9.112, §9.113). 166 PlaceReinforcements(Vec<UnitPlacement>), 167 168 // -- Scenario-specific ----- </pre>
omdurman-rules/src/effects/dispatch.rs:47 GH:omdurman-rules/src/effects/dispatch.rs:47	apply_place_reinforceme	<pre> 45 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), 46 GameEffect::PlaceReinforcements(placements) => { 47 apply_place_reinforcements(state, placements) 48 } 49 GameEffect::DervishDesertion { roll, deserters } => { </pre>
omdurman-rules/src/reinforcements.rs:69 GH:omdurman-rules/src/reinforcements.rs:69	dervish_campaign_sched	<pre> 67 /// All reinforcements enter on the west edge, south of the Khor Shambat. 68 /// Each unit pays terrain cost of the hex it enters through. 69 pub fn dervish_campaign_schedule() -> ReinforcementSchedule { 70 ReinforcementSchedule { 71 player: Player::Dervish, </pre>
omdurman-types/src/lib.rs:610 GH:omdurman-types/src/lib.rs:610	Location	<pre> 608 /// `Display` is the name as printed on the map ("Fort Buri"), for players. 609 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 610 pub enum Location { 611 #[strum(serialize = "Fort Makran")] 612 FortMakran, </pre>
omdurman-types/src/lib.rs:720 GH:omdurman-types/src/lib.rs:720	SetupLetter	<pre> 718 /// Each letter marks a specific hex where a Dervish leader is placed. 719 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 720 pub enum SetupLetter { 721 Y, 722 K, </pre>
omdurman-types/src/lib.rs:863 GH:omdurman-types/src/lib.rs:863	Faction	<pre> 861 /// `Some(BrigadeId::friendlies())`. 862 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 863 pub enum Faction { 864 Dervish { 865 tribe: DervishTribe, </pre>

Proven by: omdurman-rules::src::reinforcements::wave_for_turn_answers_exactly_on_the_printed_schedule

Covered by tests:

omdurman-app::src::tests::place_unit_translates_by_phase
omdurman-rules::src::reinforcements::dervish_schedule_has_three_waves
omdurman-rules::src::reinforcements::dervish_wave_one_has_baggaara_and_three_leaders
omdurman-rules::src::reinforcements::dervish_wave_two_has_hadendowa
omdurman-rules::src::reinforcements::dervish_wave_three_is_all_remaining
omdurman-rules::src::reinforcements::wave_for_turn_returns_correct_wave
omdurman-rules::src::effects::tests::campaign_reinforcements_gate_by_wave

omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force
omdurman-rules::src::unit_profiles::dervish_roster_matches_the_order_of_appearance
omdurman-rules::tests::campaign_order_of_battle::reinforcements_enter_through_their_entrance_areas

§9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

manual page unknown

The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:126 GH:omdurman-rules/src/reinforcements.rs:126	anglo_egyptian_campaign	124 /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit). 125 /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP). 126 pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule { 127 let free_leaders = vec![128 CampaignLeader::British(BritishLeader::Kitchener),

Proven by: omdurman-rules::src::reinforcements::wave_for_turn_answers_exactly_on_the_printed_schedule

Covered by tests: omdurman-rules::src::reinforcements::anglo_egyptian_schedule_has_four_waves
omdurman-rules::src::reinforcements::anglo_egyptian_leaders_available_each_wave
omdurman-rules::src::reinforcements::anglo_egyptian_turn_four_is_all_remaining
omdurman-rules::src::effects::tests::campaign_reinforcement_cap_and_double_entry
omdurman-rules::src::effects::tests::campaign_gunboats_quota_three_per_turn
omdurman-rules::src::effects::tests::reinforcement_rejected_onto_enemy_occupied_hex
omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force
omdurman-rules::tests::campaign_order_of_battle::reinforcements_enter_through_their_entrance_areas
omdurman-rules::tests::campaign_order_of_battle::the_anglo_egyptian_waves_follow_the_order_of_appearance

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
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omdurman-rules/src/effects/dispatch.rs:532 GH:omdurman-rules/src/effects/dispatch.rs:532	first_player	<pre> 530 531 /// The player who moves first in a scenario (\$4, 532 \$9.113, \$9.212, \$9.322). 533 pub fn first_player(scenario: Scenario) -> Player { 534 match scenario { 535 Scenario::Campaign => Player::AngloEgyptian,</pre>
omdurman-rules/src/effects/state/setup.rs:458 GH:omdurman-rules/src/effects/state/setup.rs:458	historical_set_up_area	<pre> 456 /// forbid the very hexes the scenario pins the 457 /// leaders to -- and most of 458 /// the ground around them. 459 pub fn historical_set_up_area(&self, placement: 460 &UnitPlacement) -> Result<(), RuleError> { 461 use crate::UnitIdentity; 462 if self.board.terrain.is_empty() {</pre>
omdurman-rules/src/board.rs:169 GH:omdurman-rules/src/board.rs:169	Board-Info::compute_zariba	<pre> 167 /// the open desert outside; a closed one is the 168 /// Zariba. Empty on boards 169 /// without a closed Zariba line. 170 pub fn compute_zariba(&self) -> Set<HexCoord> { 171 let is_land = 172 hex: HexCoord !matches!(173 (self.terrain_at(hex), Some(Terrain::Nile { .. }) None);</pre>

Covered by tests: [omdurman-rules::src::effects::tests::setup_is_sequential_first_side_then_second](#)
[omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units](#)
[omdurman-rules::tests::historical_setup_flow::historical_setup_completes_on_the_campaign_board](#)

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif
 - O: Osman Digna

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/setup.rs:239 GH:omdurman-rules/src/effects/state/setup.rs:239	in_deployment_zone	<pre> 237 /// areas ([`Self::campaign_set_up_area`]); 238 /// nothing for the 239 /// Anglo-Egyptians. 240 pub fn in_deployment_zone(&self, player: Player, 241 hex: HexCoord, is_boat: bool) -> bool { 242 // No board attached -> permissive (unit tests, 243 // unbound session). 244 if self.board.terrain.is_empty() {</pre>
omdurman-rules/src/unit.rs:85 GH:omdurman-rules/src/unit.rs:85	Dervish-Leader::setup_letter	<pre> 83 /// (\$9.212): A→Ali Wad Helu, D→Sheik El Din, 84 /// Y→Yakub, K→Khalifa Abdullah, 85 /// S→Sherif, O→Osman Digna. Inverse of 86 /// [`dervish_leader_for_setup_letter`]. 87 pub fn setup_letter(self) -> SetupLetter { 88 match self { 89 DervishLeader::AliWadHelu => 90 SetupLetter::A,</pre>

omdurman-rules/src/unit.rs:101 GH:omdurman-rules/src/unit.rs:101	dervish_leader_for_setup	<pre> 99 /// inherent impl here, so the mapping is a free function -- the bijective 100 /// inverse of [<code>DervishLeader::setup_letter`</code>]. 101 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 102 match letter { 103 SetupLetter::A => DervishLeader::AliWadHelu,</pre>
omdurman-rules/src/unit.rs:70 GH:omdurman-rules/src/unit.rs:70	DervishLeader::of_tribe	<pre> 68 /// The leader whose colour `tribe` wears (see [<code>Self::commands`</code>): the 69 /// leader a Historical-scenario unit sets up near (\$9.212). 70 pub fn of_tribe(tribe: DervishTribe) -> Option<DervishLeader> { 71 match tribe { 72 DervishTribe::Taiasha => Some(DervishLeader::KhalifaAbdullah),</pre>
omdurman-rules/src/effects/state/setup.rs:458 GH:omdurman-rules/src/effects/state/setup.rs:458	historical_set_up_area	<pre> 456 /// forbid the very hexes the scenario pins the leaders to -- and most of 457 /// the ground around them. 458 pub fn historical_set_up_area(&self, placement: &UnitPlacement) -> Result<(), RuleError> { 459 use crate::UnitIdentity; 460 if self.board.terrain.is_empty() {</pre>

Covered by tests:

- omdurman-rules::tests::historical_setup_flow::historical_setup_completes_on_the_campaign_board
- omdurman-rules::src::effects::tests::historical_dervish_set_up_out_of_sight_of_the_army
- omdurman-rules::src::effects::tests::deploy_rejected_outside_zone
- omdurman-rules::src::unit_profiles::embedded_leaders_resolve_from_their_host_section
- omdurman-app::src::scenario_setup::historical_places_all_six_leaders_when_anchors_present
- omdurman-app::src::scenario_setup::missing_anchor_is_reported_not_dropped_silently
- omdurman-rules::src::effects::tests::setup_letter_dervish_leader_roundtrip
- omdurman-rules::src::effects::tests::setup_letter_to_dervish_leader_known_values
- omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units

§9.231 – Thorn hedge hexsides

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:28 GH:omdurman-rules/src/combat.rs:28	ZaribaThornHedge	<pre> 26 Terrain(i16), 27 /// -2 thorn-hedge defensive modifier (§9.231). 28 ZaribaThornHedge, 29 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 30 /// units (those Nile-side of the trench hexside).</pre>
omdurman-types/src/lib.rs:277 GH:omdurman-types/src/lib.rs:277	ZaribaThornHedge	<pre> 275 Crest, 276 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 277 ZaribaThornHedge, 278 /// Historical-scenario trench segment of the Zariba (§9.232). 279 ZaribaTrench,</pre>
omdurman-rules/src/effects/state.rs:349 GH:omdurman-rules/src/effects/state.rs:349	has_zariba_thorn_hedge	<pre> 347 /// *effective* hexsides, so constructed (§5.3) and authored hedges both 348 /// count. 349 pub fn has_zariba_thorn_hedge(&self, hex: HexCoord) -> bool { 350 for n in hex.neighbors() { 351 if self.hexside_effective_is(hex, n, k k == HexsideKind::ZaribaThornHedge) {</pre>

Covered by tests: `omdurman-rules::src::effects::tests::zariba_fire_penalties_apply_to_dervish_fire_only`
`omdurman-rules::src::effects::tests::the_hedge_penalises_fire_across_it_only`

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/combat.rs:31 GH:omdurman-rules/src/combat.rs:31	ZaribaTrenchEntrenched	<pre>29 /// -4 trench defensive modifier (§9.232). Only 30 applies vs. "entrenched" 31 /// units (those Nile-side of the trench hexside). 32 ZaribaTrenchEntrenched, 33 }</pre>
omdurman-types/src/lib.rs:279 GH:omdurman-types/src/lib.rs:279	ZaribaTrench	<pre>277 ZaribaThornHedge, 278 /// Historical-scenario trench segment of the 279 Zariba (§9.232). 280 ZaribaTrench, 281 /// One of the two end hexsides of a Zariba trench 282 segment that connect to 283 /// the Nile River (§9.233). Units may only enter/ 284 leave the Zariba via</pre>
omdurman-rules/src/effects/state.rs:337 GH:omdurman-rules/src/effects/state.rs:337	is_zariba_entrenched	<pre>335 /// the hex across the trench is the open desert, 336 never the river. Reads 337 /// *effective* hexsides. 338 pub fn is_zariba_entrenched(&self, hex: HexCoord) - 339 > bool { 340 self.board.is_zariba(hex) 341 && hex</pre>

Proven by: `omdurman-rules::src::verification::melee_modifier_keeps_roll_legal`

Covered by tests: `omdurman-rules::src::effects::tests::trench_entrenched_units_take_trench_modifiers`
`omdurman-rules::src::effects::tests::entrenched_units_do_not_block_line_of_sight`

§9.233 – Zariba entry/exit costs

implemented

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:336 GH:omdurman-types/src/lib.rs:336	blocks_movement	<pre>334 /// `omdurman-rules`). The trench *end* variants 335 are therefore intentionally 336 /// not blocking. 337 pub fn blocks_movement(self) -> bool { 338 matches!(339 self,</pre>
omdurman-rules/src/terrain_chart.rs:100 GH:omdurman-rules/src/terrain_chart.rs:100	hexside_movement_surcl	<pre>98 /// (a Gate or Breach costs +1; the Wall itself is 99 closed), and the §9.233 100 /// +2 MP Zariba trench ends (the rest of the Zariba is 101 closed, §9.23). 102 pub fn hexside_movement_surcharge(hexside: 103 Option<HexsideKind>) -> Option<i16> { 104 use HexsideKind::*; 105 match hexside {</pre>
omdurman-rules/src/ef-fects/state/	movement_cost_for	<pre>542 /// 543 /// §5.42: entering or leaving an enemy ZOC adds no</pre>

movement.rs:544 GH:omdurman-rules/src/effects/state/movement.rs:544		MP cost. 544 pub fn movement_cost_for(545 &self, 546 unit: &UnitPlacement,
omdurman-rules/src/effects/movement.rs:9 GH:omdurman-rules/src/effects/movement.rs:9	apply_move_unit	7 /// engine from the board (\$5.11/\$5.24): the caller-supplied `cost` is ignored 8 /// (it survives in the effect only for wire compatibility). 9 pub fn apply_move_unit(10 state: &mut GameState, 11 unit_id: UnitId,

Proven by: [omdurman-types::src::lib::zariba_trench_ends_differ_only_in_the_entry_rule](#)

Covered by tests: [omdurman-rules::src::effects::tests::zariba_end_hexside_costs_extra_mp](#)
[omdurman-rules::src::effects::tests::zariba_thorn_hedge_blocks_movement](#)
[omdurman-app::src::picker::movement::trench_end_crossing_costs_two_extra_mp](#)

§9.321 – British set up (Bonus)

[implemented](#)

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
 - one Egyptian Battalion artillery unit
 - two British infantry units (represents Caucasian troops)
 - three Egyptian infantry units (represents Cairo “Bazouks”)
 - four Sudan infantry units (represents Sudanese blacks)
 - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:45 GH:omdurman-rules/src/scenario_setup.rs:45	FALL_OF_KHARTOUM	43 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 44 /// sprites). 45 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[46 FixedPlacement { 47 section: SectionName::BritishBoats,

Covered by tests: [omdurman-rules::src::effects::tests::setup_is_sequential_first_side_then_second](#)
[omdurman-rules::src::effects::tests::printed_fok_forts_are_british_forts](#)
[omdurman-rules::src::effects::tests::fall_of_khartoum_has_no_maxim_howitzzer_subphase](#)
[omdurman-rules::src::effects::tests::confirm_ready_rejected_below_scenario_target](#)
[omdurman-rules::src::effects::tests::remove_deployed_unit_happy_path](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_reports_missing_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
[omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile](#)
[omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob](#)
[omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine](#)
[omdurman-rules::src::effects::tests::british_boats_named_vs_old_gunboat_detection](#)
[omdurman-rules::src::effects::tests::fok_order_of_battle_british](#) [omdurman-rules::src::effects::tests::fok_order_of_battle_dervish](#)

omdurman-rules::src::effects::tests::fok_caps_bind_across_counter_variants
omdurman-rules::src::effects::tests::fok_ae_may_set_up_behind_a_gate

§9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi's combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:398 GH:omdurman-rules/src/unit_profiles.rs:398	ali_wad_helu	<pre>396 /// §9.2/§9.3) unplaceable -- the FoK order-of-battle table 397 /// (`fok_cap_group`) is keyed by identity. 398 pub fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 399 match (col, row) { 400 (0, 0) => dervish_leader(DervishLeader::AliWadHelu),</pre>
omdurman-types/src/lib.rs:971 GH:omdurman-types/src/lib.rs:971	sections_for_picker	<pre>969 /// counter of those two forces and leave the §9.322 entry force -- and 970 /// with it the setup Ready gate (§9.2/§9.3) -- unplaceable. 971 pub fn sections_for_picker(self) -> Option<&'static [SectionName]> { 972 match self { 973 Scenario::Campaign Scenario::Historical => None,</pre>

Covered by tests: omdurman-rules::src::effects::tests::deployment_requires_the_canonical_counter
omdurman-rules::src::effects::tests::fok_has_no_reinforcements
omdurman-rules::src::unit_profiles::ali_wad_helu_block_resolves_leader_and_degelim_tribes
omdurman-rules::tests::fok_setup_flow::degheim_counters_resolve_to_the_degheim_tribe
omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob
omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile
omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks
omdurman-rules::src::effects::tests::fok_order_of_battle_dervish
omdurman-rules::src::effects::tests::fok_dervish_east_edge_on_diamond_board
omdurman-rules::src::effects::tests::fok_dervish_pay_their_entry_hex_on_turn_one

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:169 GH:omdurman-rules/src/turn_track.rs:169	FALL_OF_KHARTOUM	<pre>167 /// rule-bearing (night halves Anglo-Egyptian movement and ranges and bars 168 /// howitzer fire, §8.1). 169 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [</pre>

		170 TurnEntry { 171 turn: 1,
--	--	---

Proven by: [omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turn_one](#)

Covered by tests: [omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_turns_2_to_8_are_day](#)

§9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	fall_of_khartoum_map	35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData { 38 FOK.get_or_init(parse(FOK_RON)).clone() 39 }

Covered by tests: [omdurman-rules::src::effects::tests::fall_of_khartoum_board_excludes_no_hexes](#)

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/fire.rs:85 GH:omdurman-rules/src/effects/fire.rs:85	range_band_for	83 /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table 84 /// (§9.343). 85 pub fn range_band_for(86 scenario: Scenario, 87 player: Player,

Proven by: [omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table](#)
[omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlyes_and_fok_to_the_dervish_table](#)

Covered by tests: [omdurman-rules::src::effects::tests::fok_both_players_use_dervish_range_table](#)

§9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:45	FALL_OF_KHARTOUM	43 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 44 /// sprites).

GH:omdurman-rules/src/scenario_setup.rs:45		<pre> 45 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = 46 &[47 FixedPlacement { section: SectionName::BritishBoats, }] </pre>
omdurman-rules/src/effects/state.rs:428 GH:omdurman-rules/src/effects/state.rs:428	hex_has_enemy_fort	<pre> 426 /// the British Forts Makran and Buri (§9.321) 427 included. A destroyed fort 428 /// leaves only the printed building behind. 429 pub fn hex_has_enemy_fort(&self, hex: HexCoord, 430 mover: Player) -> bool { 431 self.units.iter().any(u { 432 u.position == hex 433 }) 434 } </pre>

Covered by tests: [omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
[omdurman-app::src::scenario_setup::placement_done_gate_matches_by_identity_not_allocated_id](#)
[omdurman-rules::src::effects::tests::fok_order_of_battle_dervish](#)
[omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks](#)

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:408 GH:omdurman-rules/src/effects/state.rs:408	is_nile_mouth_crossing	<pre> 406 /// must be named on the board, else this is 407 `false` and the move falls 408 /// through to the ordinary contiguous-Nile rules. 409 pub fn is_nile_mouth_crossing(&self, from: 410 HexCoord, to: HexCoord) -> bool { 411 let white = self 412 .board </pre>
omdurman-types/src/lib.rs:635 GH:omdurman-types/src/lib.rs:635	Location::WhiteNileMouth	<pre> 633 /// a British gunboat may cross to the Blue Nile 634 mouth for 6 upstream MP. 635 #[strum(serialize = "White Nile mouth")] 636 WhiteNileMouth, 637 /// The off-board mouth of the Blue Nile branch 638 (FALL OF KHARTOUM §9.345). 639 #[strum(serialize = "Blue Nile mouth")] </pre>

Covered by tests: [omdurman-rules::src::effects::tests::fok_gunboat_crosses_between_nile_mouths](#)

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:676 GH:omdurman-rules/src/effects/dispatch.rs:676	check_gordon_palace	<pre> 674 /// (overrun in passing, §6.51). A no-op outside FoK or 675 while he lives 676 /// unthreatened. 677 pub fn check_gordon_palace(state: &mut GameState) { 678 if state.scenario != Scenario::FallOfKhartoum { 679 return; 680 } </pre>

omdurman-rules/src/unit.rs:354 GH:omdurman-rules/src/unit.rs:354	UnitIdentity::is_gordon	<pre>352 /// Whether this is the GORDON leader unit (§9.32, 353 §9.346) -- the immobile 354 /// palace defender whose elimination ends FALL OF 355 KHARTOUM (§9.35). 356 pub fn is_gordon(&self) -> bool { 357 matches!(358 self,</pre>
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Covered by tests:

- omdurman-rules::src::effects::tests::breach_casualty_stands_at_the_wall
- omdurman-rules::src::unit_profiles::gordon_is_an_immobile_british_leader
- omdurman-rules::src::effects::tests::gordon_survives_means_no_elimination
- omdurman-rules::src::effects::tests::gordon_dying_in_melee_ends_fall_of_khartoum
- omdurman-rules::src::effects::tests::fire_never_kills_gordon
- omdurman-rules::src::effects::tests::a_lone_gordon_makes_no_melee_roll
- omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace

7/10 implemented

§10 – Optional Rules

§10 – Optional Rules

descriptive

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

§10.1 – River Mines

descriptive

manual page unknown

River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

§10.2 – River Chain

descriptive

manual page unknown

River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E-W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/setup.rs:328 GH:omdurman-rules/src/effects/setup.rs:328	apply_place_mine	<pre>326 /// Lay a river mine during setup (§10.11). Validated by 327 /// [<code>GameState::can_place_mine`</code>]. 328 pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> { 329 state.can_place_mine(hex)?; 330 state.mines.push(MinePlacement {</pre>
omdurman-rules/src/effects/state.rs:230 GH:omdurman-rules/src/effects/state.rs:230	GameState::mines	<pre>228 229 friendlies_transport: None, 230 optional_rules: Vec::new(), 231 mines: Vec::new(), 232 chain: None, 233 pending_mine: None,</pre>
omdurman-rules/src/	OptionalRule	<pre>56 /// two should be in play (rulebook §10). 57 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

turn.rs:58 GH:omdurman-rules/src/turn.rs:58	<pre>PartialEq, Eq, Debug)] 58 pub enum OptionalRule { 59 RiverMines, 60 RiverChain,</pre>
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Covered by tests: [omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup](#)
[omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule](#)
[omdurman-rules::src::effects::tests::river_obstacles_lie_south_of_the_khor_shambat](#)

§10.12 – Mine resolution

implemented

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When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine’s blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:185 GH:omdurman-rules/src/effects/effect.rs:185	RiverMine	<pre>183 /// River mine resolution (rulebook §10.12): the Dervish player's roll for 184 /// the British gunboat stopped on his mine (`pending_mine`). 185 RiverMine { 186 gunboat_id: UnitId, 187 hex: HexCoord,</pre>
omdurman-rules/src/effects/river.rs:127 GH:omdurman-rules/src/effects/river.rs:127	apply_river_mine	<pre>125 /// gunboat drifts from now on), 8-10 sunk. Only the pending strike may be 126 /// resolved, and each mine goes off once (§10.13). 127 pub fn apply_river_mine(128 state: &mut GameState, 129 gunboat_id: UnitId,</pre>
omdurman-rules/src/transport.rs:51 GH:omdurman-rules/src/transport.rs:51	MineResult	<pre>49 /// British gunboat enters a mined hex. 50 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 51 pub enum MineResult { 52 /// Roll 1-4: no effect. 53 NoEffect,</pre>

Proven by: [omdurman-rules::src::verification::mine_result_bands_match_the_printed_rule](#)
[omdurman-rules::src::effects::river::river_mine_triggers_even_on_a_harmless_roll](#)
[omdurman-rules::src::effects::river::river_mine_engine_loss_arms_the_drift_flag](#)
[omdurman-rules::src::effects::river::river_mine_sinking_removes_the_gunboat](#)

Covered by tests: [omdurman-rules::src::effects::tests::a_mined_gunboat_scores_and_takes_its_friendlys_down](#)
[omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)
[omdurman-rules::src::effects::tests::a_disabled_gunboat_drifts_two_hexes_a_turn](#)

§10.13 – Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:127 GH:omdurman-rules/src/effects/river.rs:127	apply_river_mine	<pre> 125 /// gunboat drifts from now on), 8-10 sunk. Only the pending strike may be 126 /// resolved, and each mine goes off once (§10.13). 127 pub fn apply_river_mine(state: &mut GameState, 128 gunboat_id: UnitId, 129 </pre>

Proven by: [omdurman-rules::src::effects::river::river_mine_triggers_even_on_a_harmless_roll](#)
[omdurman-rules::src::effects::river::a_triggered_mine_never_fires_again](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)

§10.14 – Dervish gunboats pass safely

implemented

manual page unknown

The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:127 GH:omdurman-rules/src/effects/river.rs:127	apply_river_mine	<pre> 125 /// gunboat drifts from now on), 8-10 sunk. Only the pending strike may be 126 /// resolved, and each mine goes off once (§10.13). 127 pub fn apply_river_mine(state: &mut GameState, 128 gunboat_id: UnitId, 129 </pre>

Proven by: [omdurman-rules::src::effects::river::dervish_gunboats_pass_mined_hexes_unharmed](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish](#)

§10.21 – Secretly record chain hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/setup.rs:809 GH:omdurman-rules/src/effects/state/setup.rs:809	can_place_chain	<pre> 807 /// Read-only check of a river-chain placement in setup (§10.21): Setup phase 808 /// and at most [<code>MAX_CHAIN_HEXES</code>] hexes. 809 pub fn can_place_chain(&self, hexes: &[HexCoord]) - > Result<(), RuleError> { 810 self.require_setup_phase()?; 811 // §10: "Optional Rules (Campaign game only)". </pre>
omdurman-rules/src/effects/state/setup.rs:339 GH:omdurman-rules/src/effects/state/setup.rs:339	apply_place_chain	<pre> 337 /// Lay (or replace) the river chain during setup (§10.21). Validated by 338 /// [<code>GameState::can_place_chain</code>]. 339 pub fn apply_place_chain(state: &mut GameState, hexes: &[HexCoord]) -> Result<(), RuleError> { 340 state.can_place_chain(hexes)?; 341 state.chain = Some(ChainPlacement { </pre>
omdurman-rules/src/effects/state/	MAX_CHAIN_HEXES	<pre> 130 131 /// Maximum contiguous Nile hexes the river chain may span (§10.21). </pre>

setup.rs:132 GH:omdurman-rules/src/effects/state/setup.rs:132		<pre> 132 pub const MAX_CHAIN_HEXES: usize = 4; 133 134 impl GameState { </pre>
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Covered by tests: [omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup](#)
[omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule](#)
[omdurman-rules::src::effects::tests::river_obstacles_lie_south_of_the_khor_shambat](#)

§10.22 – Gunboat stops on chained hex

implemented

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state/movement.rs:575 GH:omdurman-rules/src/effects/state/movement.rs:575	can_move_gunboat	<pre> 573 /// The engine computes the cost (one MP per hex, a 574 /// flat 6 for the \$9.345 575 /// mouth crossing); a larger caller `cost` must 576 /// also fit. 577 pub fn can_move_gunboat(578 &self, 579 unit_id: UnitId, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk](#)

§10.23 – Sinking the chain

implemented

manual page unknown

No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:181 GH:omdurman-rules/src/effects/river.rs:181	apply_sink_chain	<pre> 179 /// its own range, are summed onto one Combat Results 180 /// Table row, and a 181 /// result of 3 or more sinks the chain. 182 pub fn apply_sink_chain(183 state: &mut GameState, 184 firers: &[UnitId], </pre>

Covered by tests: [omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk](#)
[omdurman-rules::src::effects::tests::a_battalion_on_the_bank_sinks_the_chain](#)
[omdurman-rules::src::effects::tests::artillery_sinks_the_chain_on_three_or_more](#)

0/1 implemented

§11 – Historical Notes

§11 – Historical Notes

descriptive

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Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

0/1 implemented

Credits

§Credits – Credits

descriptive

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 1/1 implemented

Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:10 GH:omdurman-rules/src/combat_results_table.rs:10	FireFactorRow	<pre>8 serde::Serialize, serde::Deserialize, Clone, Copy, 9 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 10 pub enum FireFactorRow { 11 /// 1-5 factors 12 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:61 GH:omdurman-rules/src/combat_results_table.rs:61	from_total	<pre>59 60 /// Determine which row a given total fire factor 61 falls into (rulebook §6.22). 62 pub fn from_total(total: u16) -> Self { 63 match total { 64 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:103 GH:omdurman-rules/src/combat_results_table.rs:103	combat_results_table	<pre>101 /// D = `Disrupt` (1/2 of target units, round up) 102 /// 1...5 = `Eliminate(n)` (that many units removed) 103 pub fn combat_results_table(row: FireFactorRow, roll: 104 DieRoll) -> CombatResult { 105 crate::tables_data::CRT[row.index()][(roll.value() 106 - 1) as usize]</pre>
omdurman-rules/src/combat.rs:104 GH:omdurman-rules/src/combat.rs:104	CombatResult	<pre>102 /// * `--` -- no effect 103 #[derive(Serialize, Deserialize, Clone, Copy, 104 PartialEq, Eq, Debug)] 105 pub enum CombatResult { 106 NoEffect, 107 Disrupt,</pre>

Proven by: [omdurman-rules::src::verification::disrupt_half_is_rounded_up](#)
[omdurman-rules::src::combat_results_table::crt_eliminate_count_stays_within_printed_bounds](#)
[omdurman-rules::src::combat_results_table::crt_is_monotone_in_the_die_roll_for_every_row](#)

Covered by tests: [omdurman-rules::src::combat_results_table::ae_combat_results_table_lowest_is_no_effect](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_highest_is_eliminate_5](#)
[omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_roll](#)
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[omdurman-rules::src::combat_results_table::fire_factor_row_boundaries](#)
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[omdurman-rules::src::combat_results_table::fire_factor_row_index_sequential](#)
[omdurman-rules::src::combat_results_table::crt_all_rows_monotone_non_decreasing](#)
[omdurman-rules::src::combat_results_table::crt_cross_row_monotone_for_each_roll](#)
[omdurman-rules::src::combat_results_table::crt_lowest_row_is_worst_highest_row_is_best](#)
[omdurman-rules::src::combat_results_table::crt_eliminate_never_exceeds_5](#)
[omdurman-rules::src::combat_results_table::crt_every_cell_matches_the_table](#)

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| Reference – Charts and Tables

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